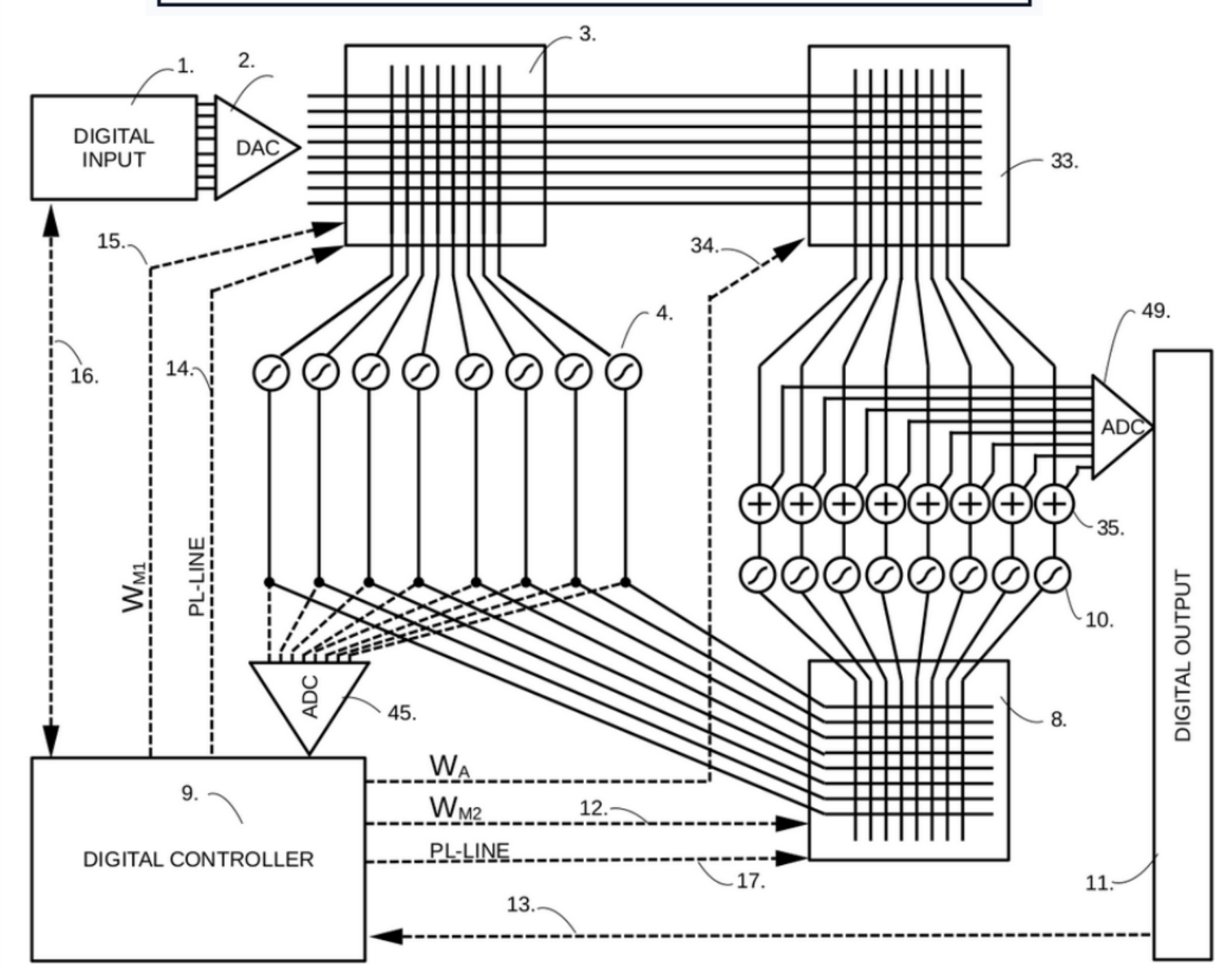
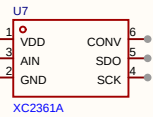
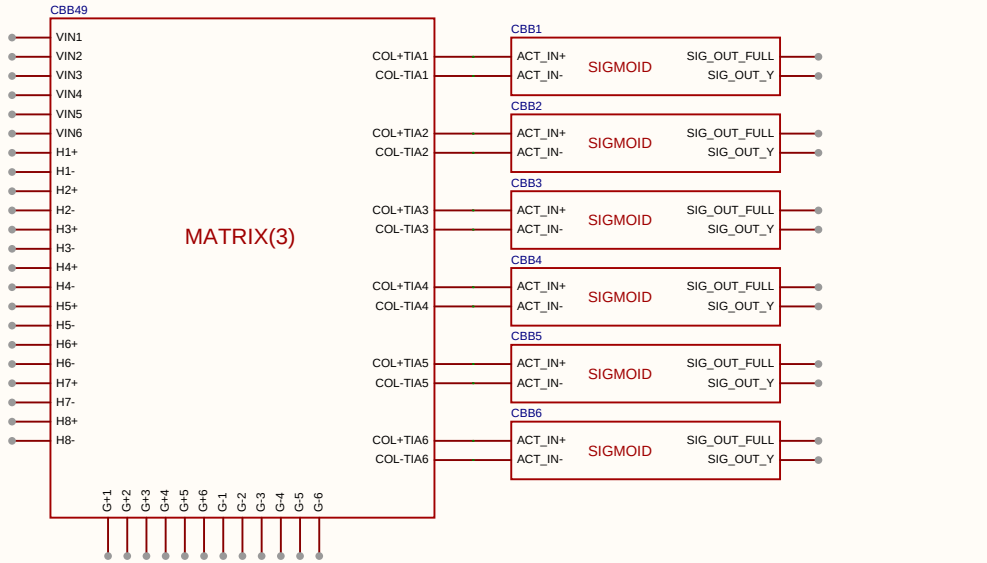
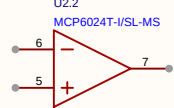
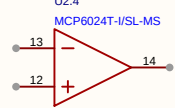
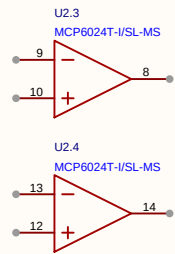
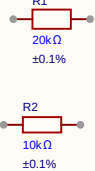
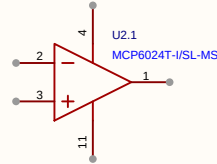
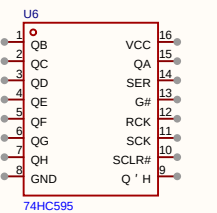
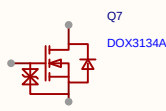


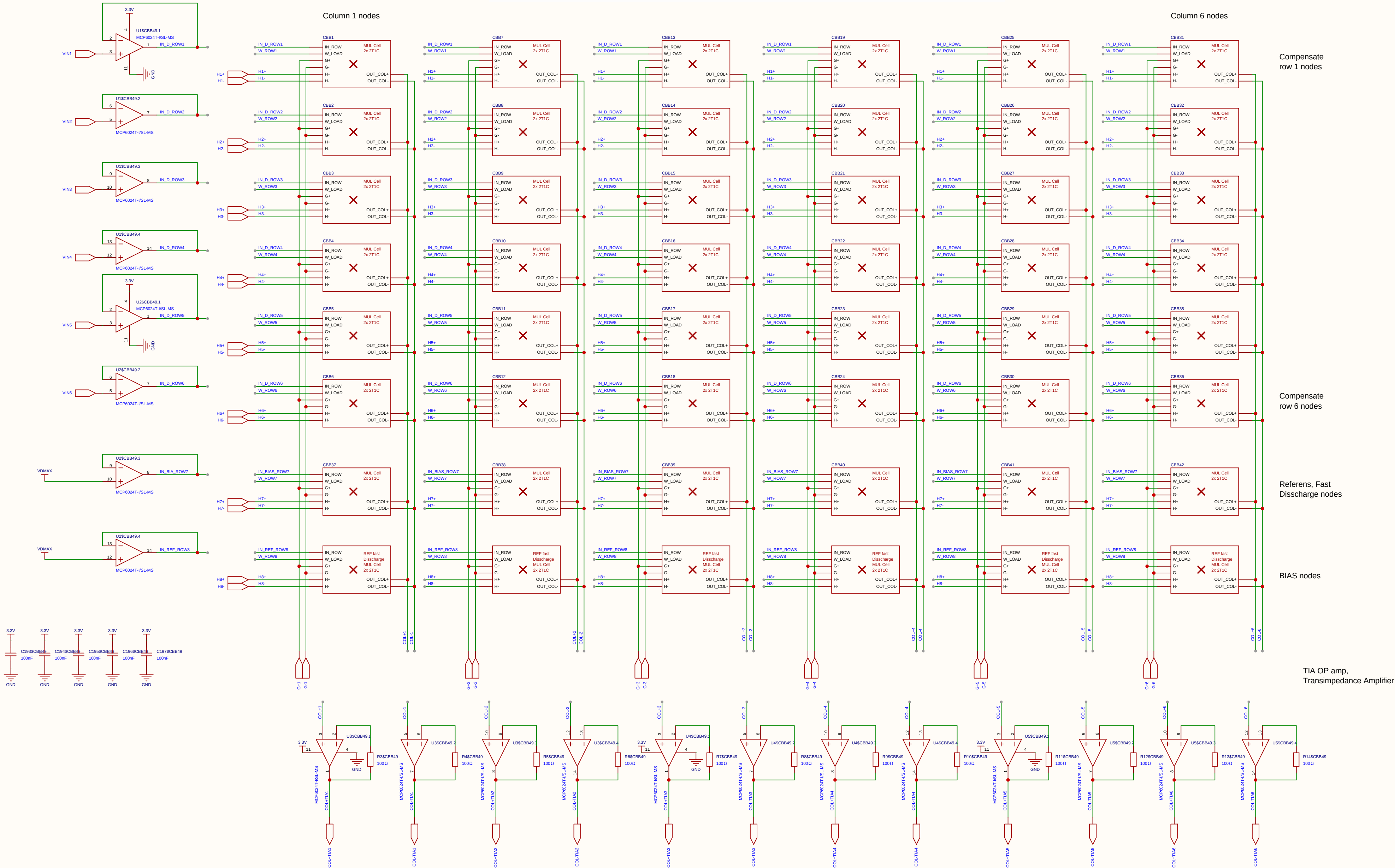
TITLE: INVERTED MAML FOR ANALOG IMC MESH
PATENT APP NO: SE 2630397-4
STATUS: PATENT PENDING
INVENTOR: OLLE WELIN
PRIORITY DATE: 2026-06-03



Schematic	Inverted_MAML			Create at	2026-06-10
				Update at	2026-06-10
Board				Page	cover
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 2	
		V1.0	A4	EasyEDA.com	



$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3			Create at	2026-06-14
				Update at	2026-06-14
Board				Page	matx3
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

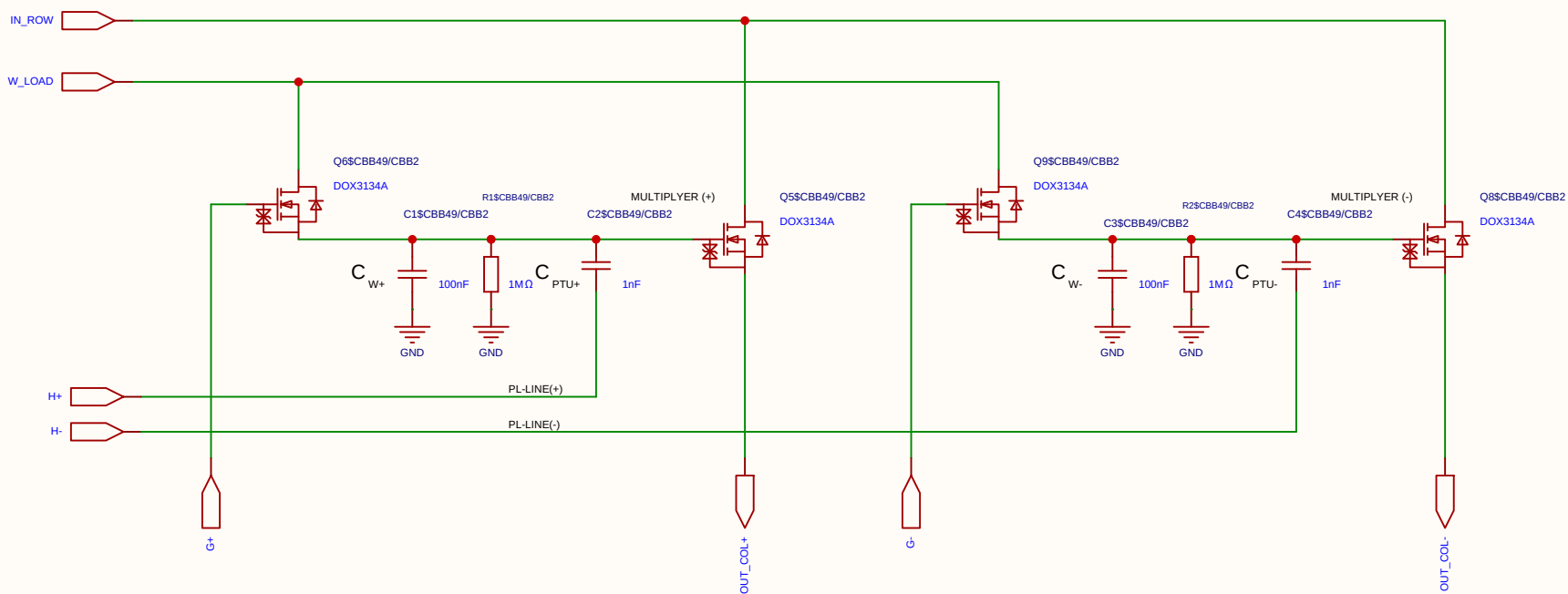
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

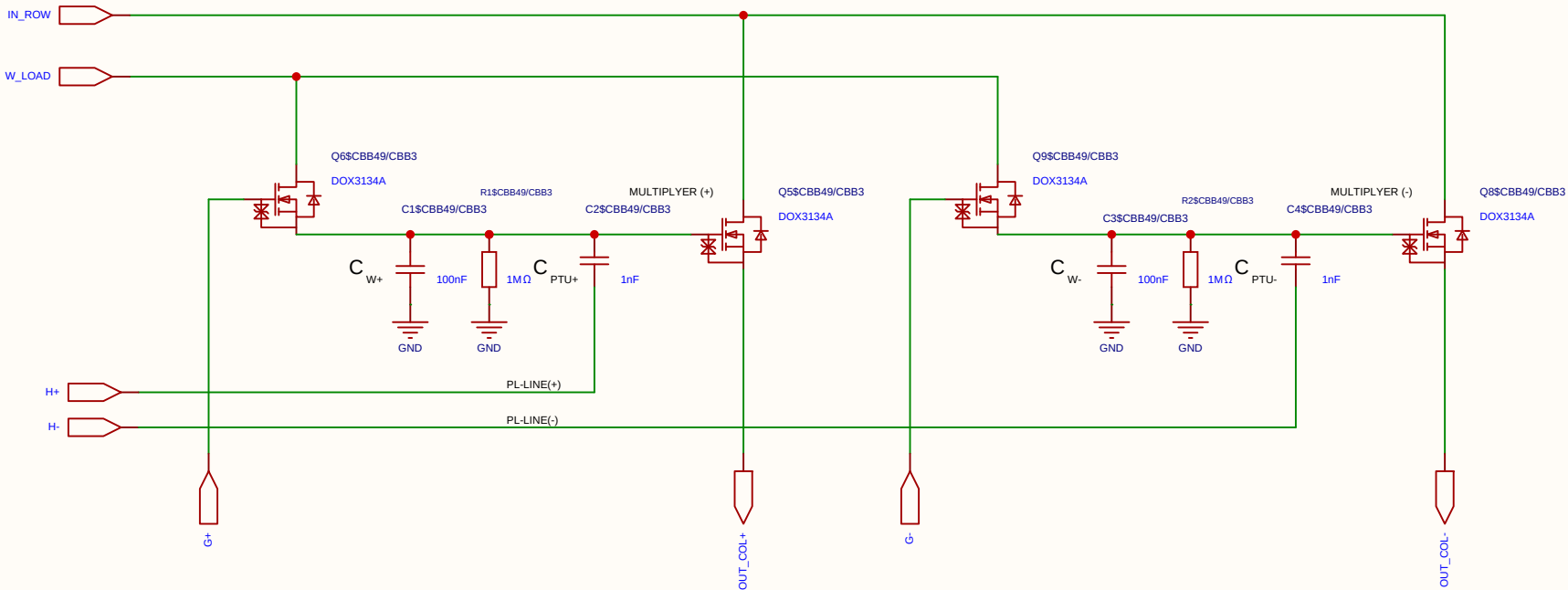
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

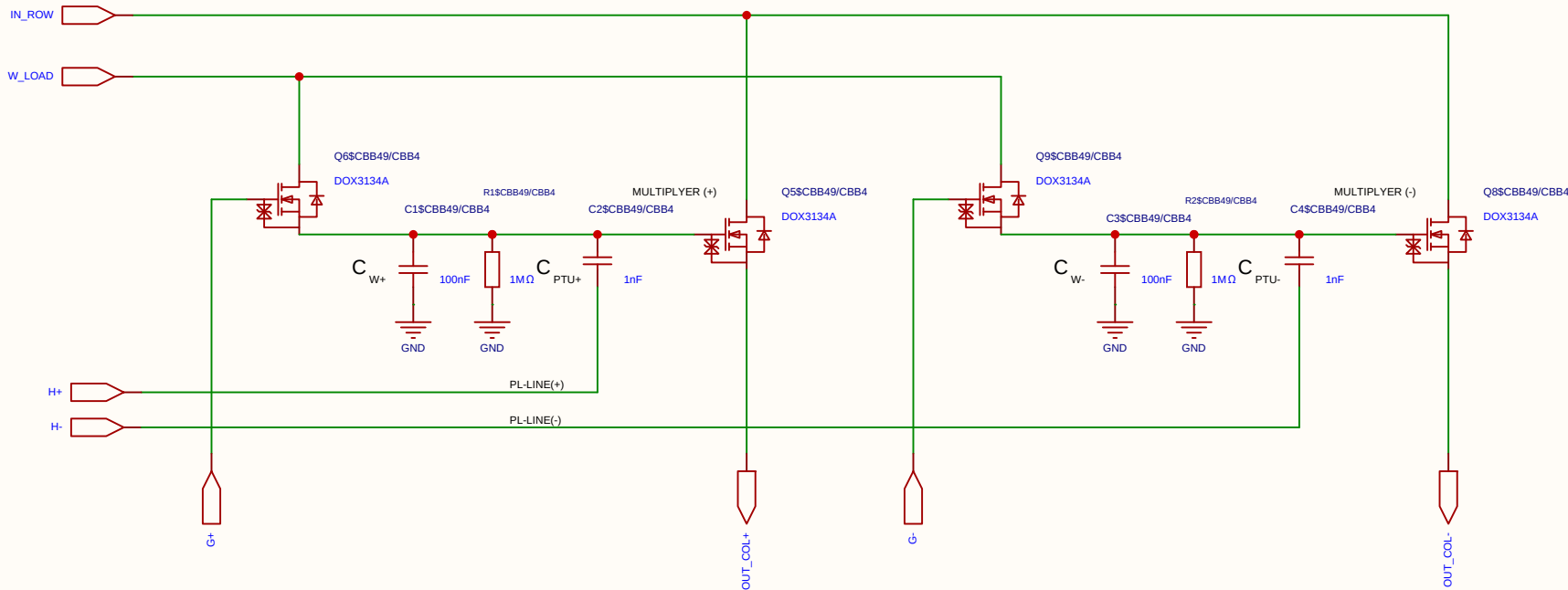
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

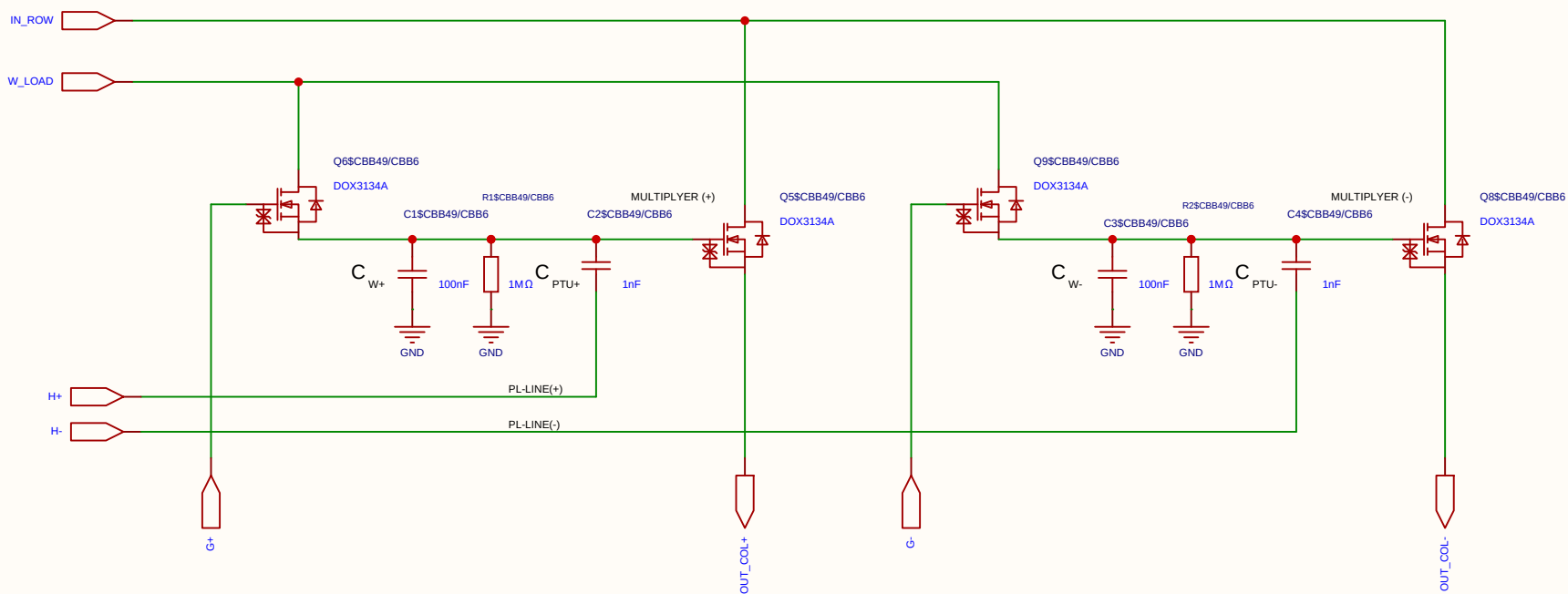
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

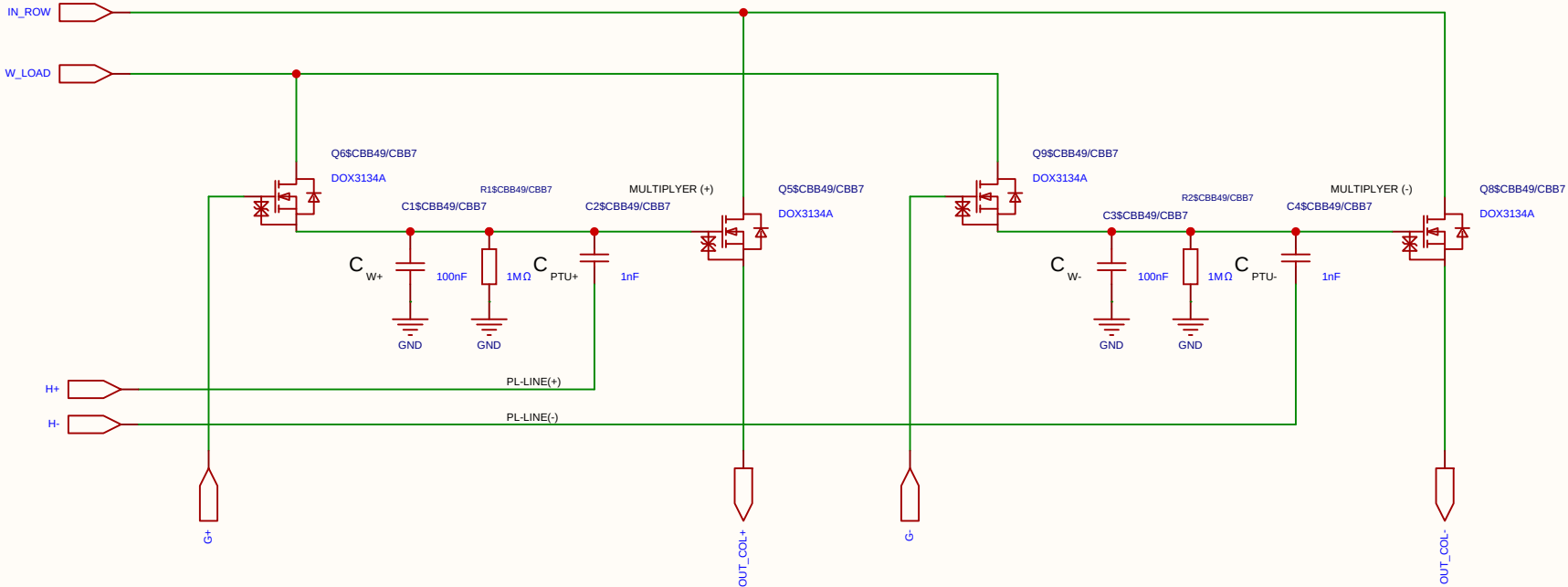
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

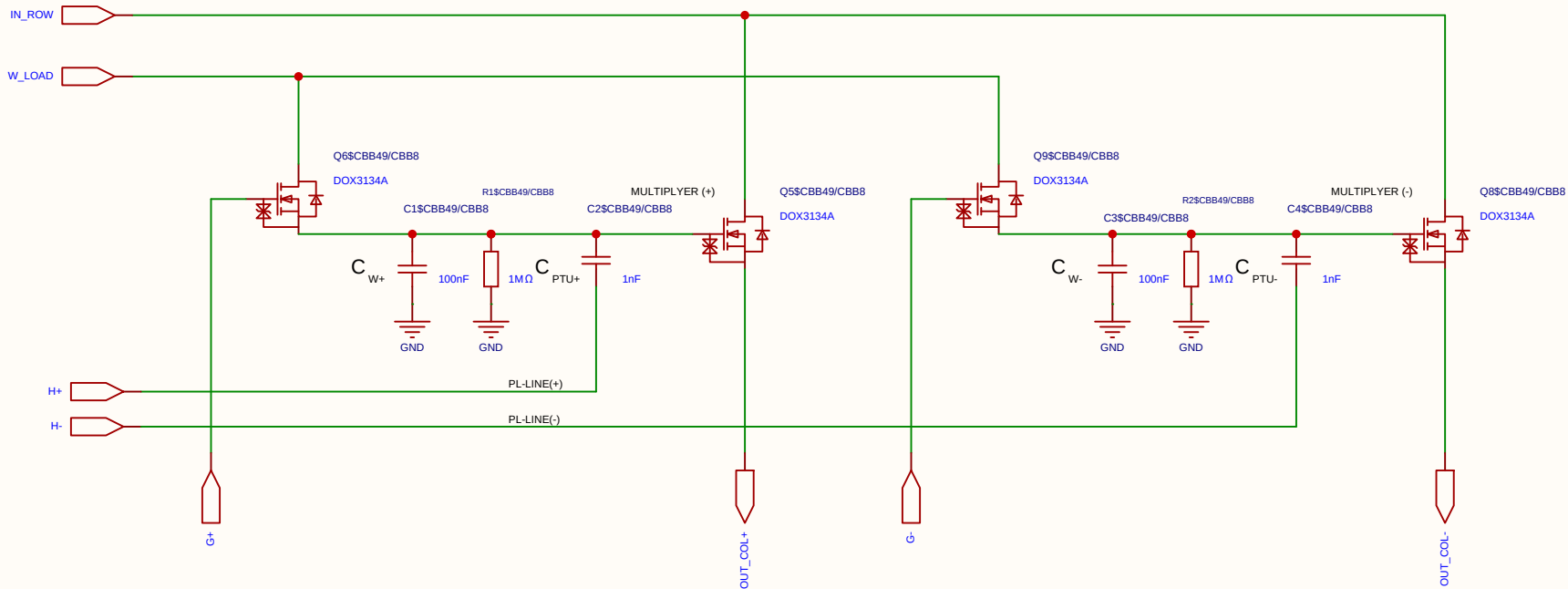
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

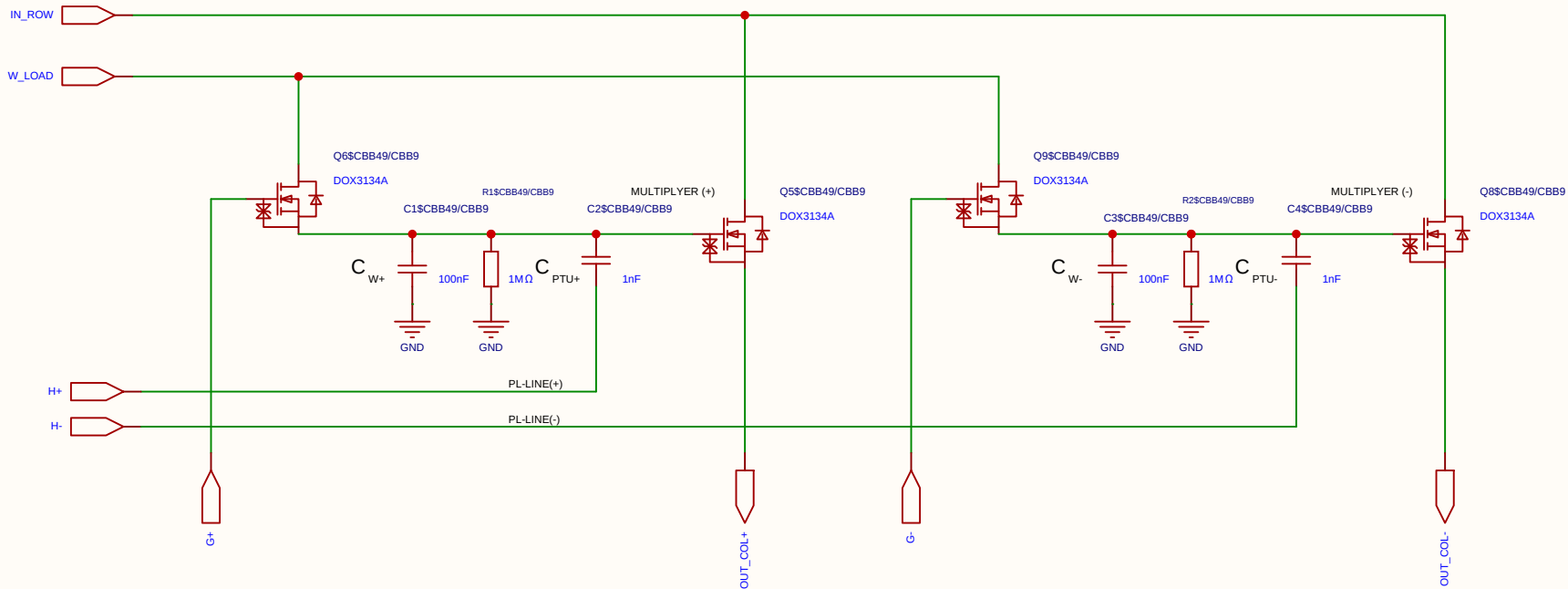
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

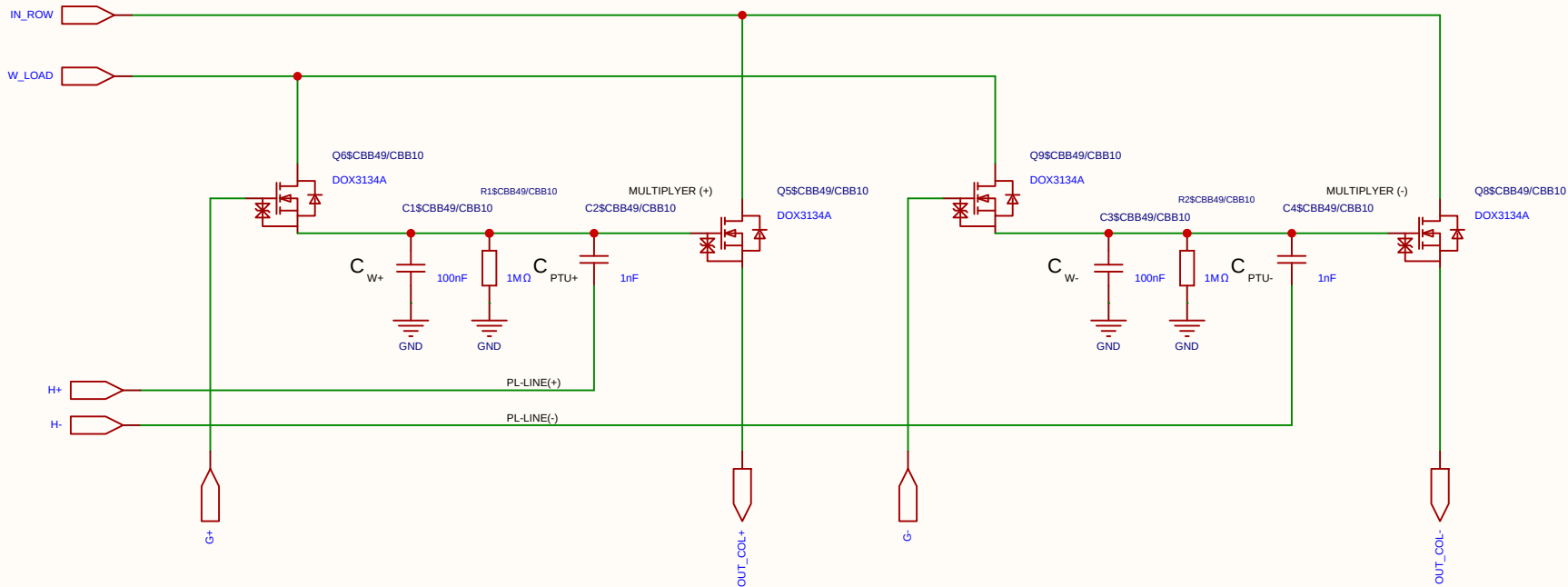
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

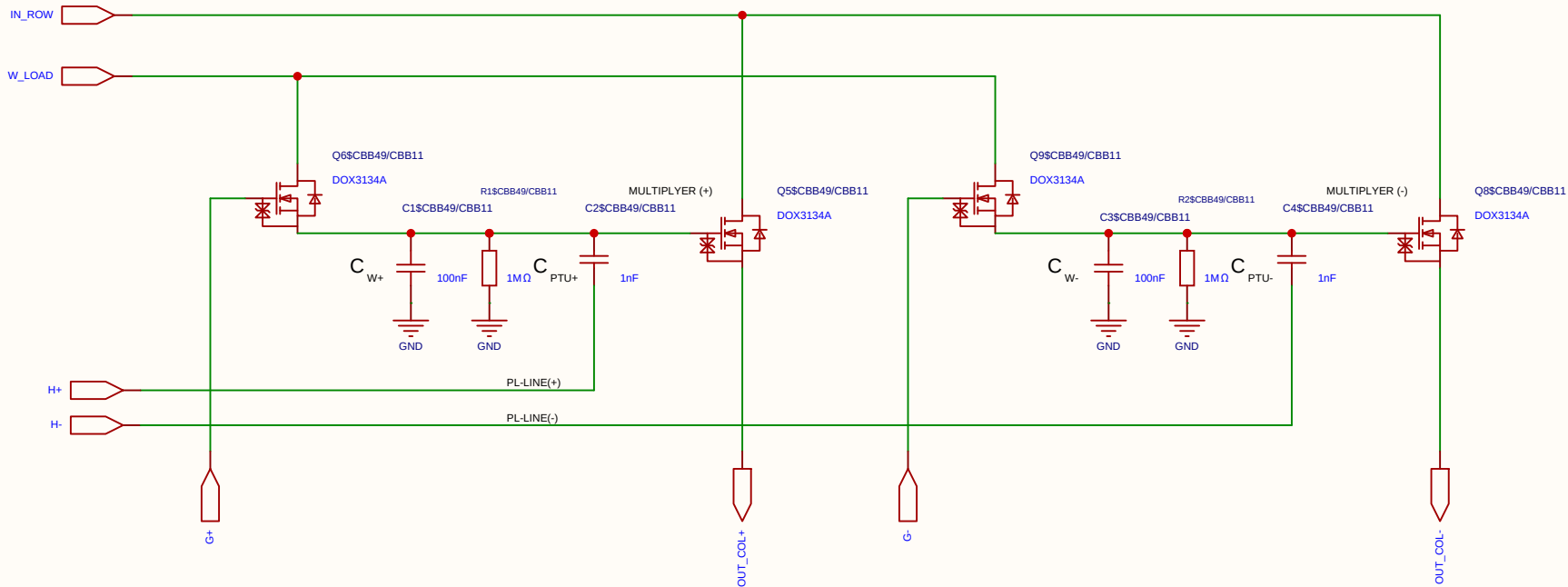
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

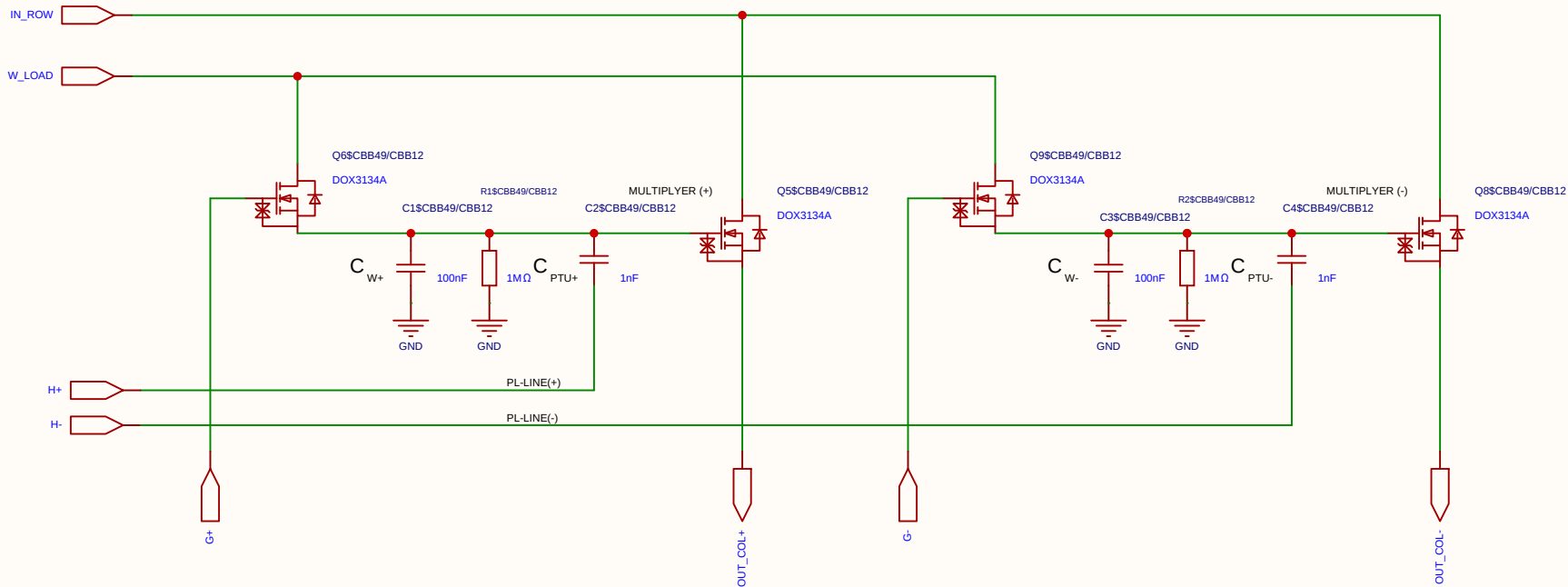
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

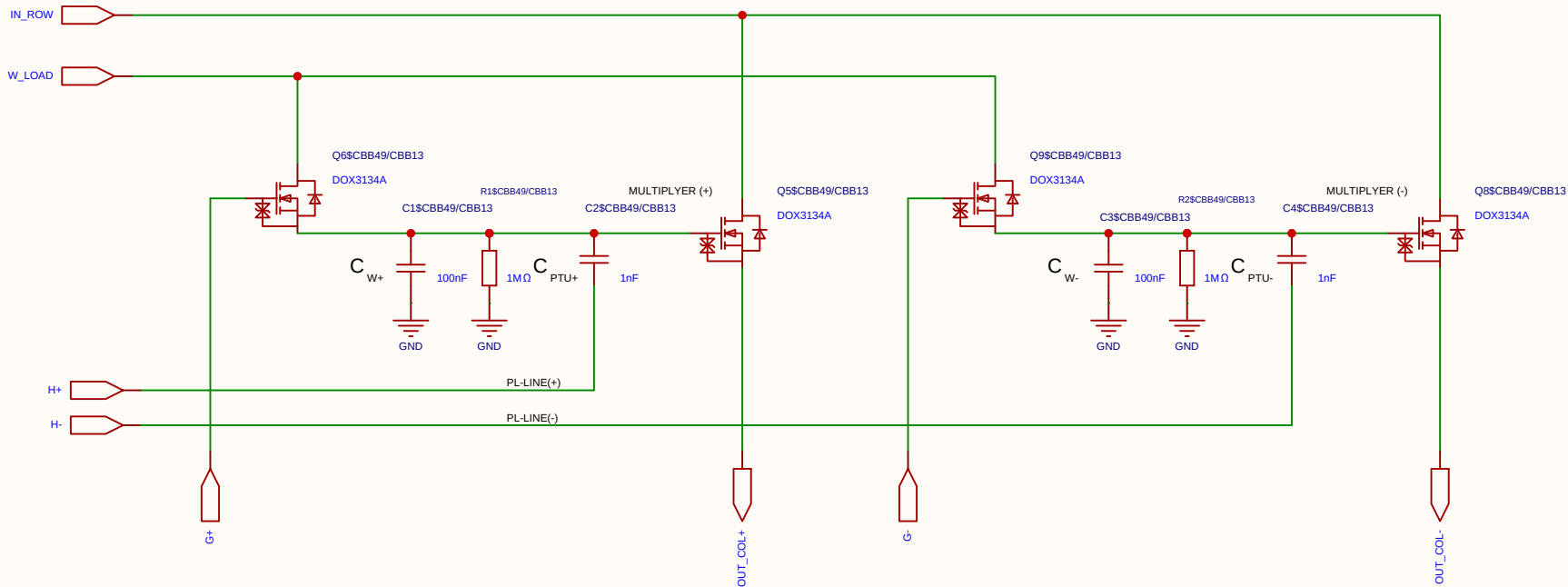
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

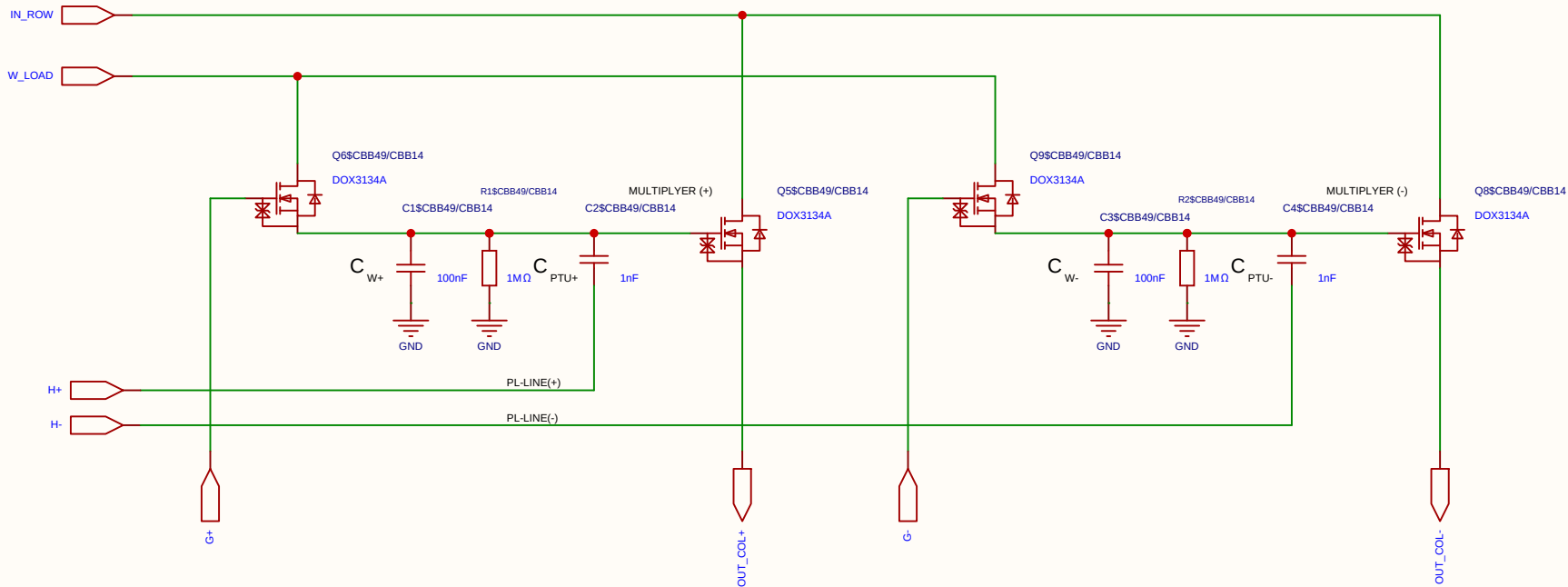
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

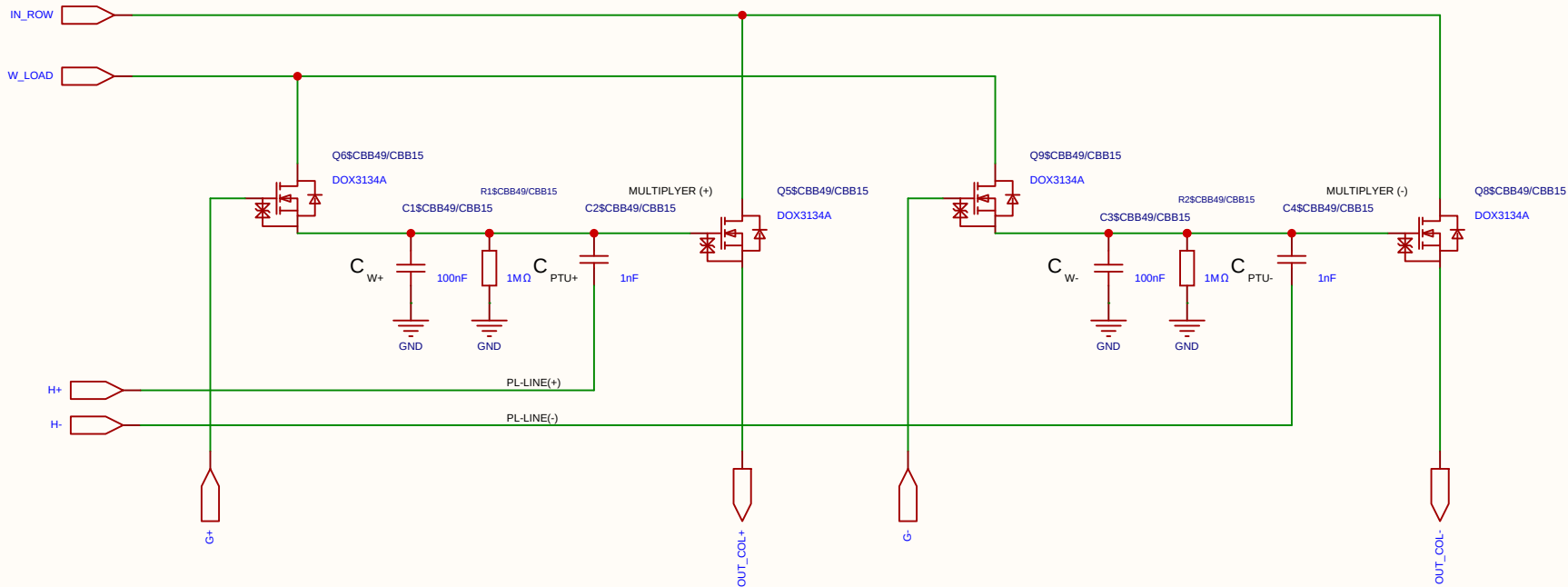
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
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Drawn		Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

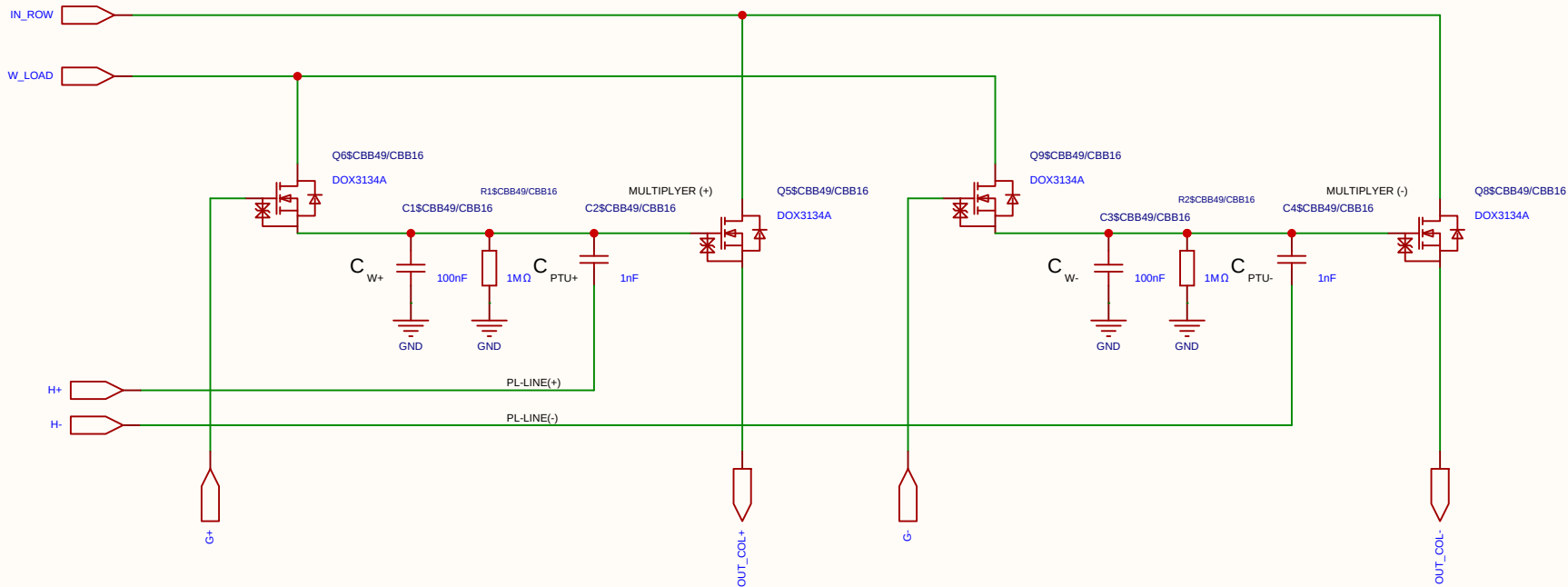
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

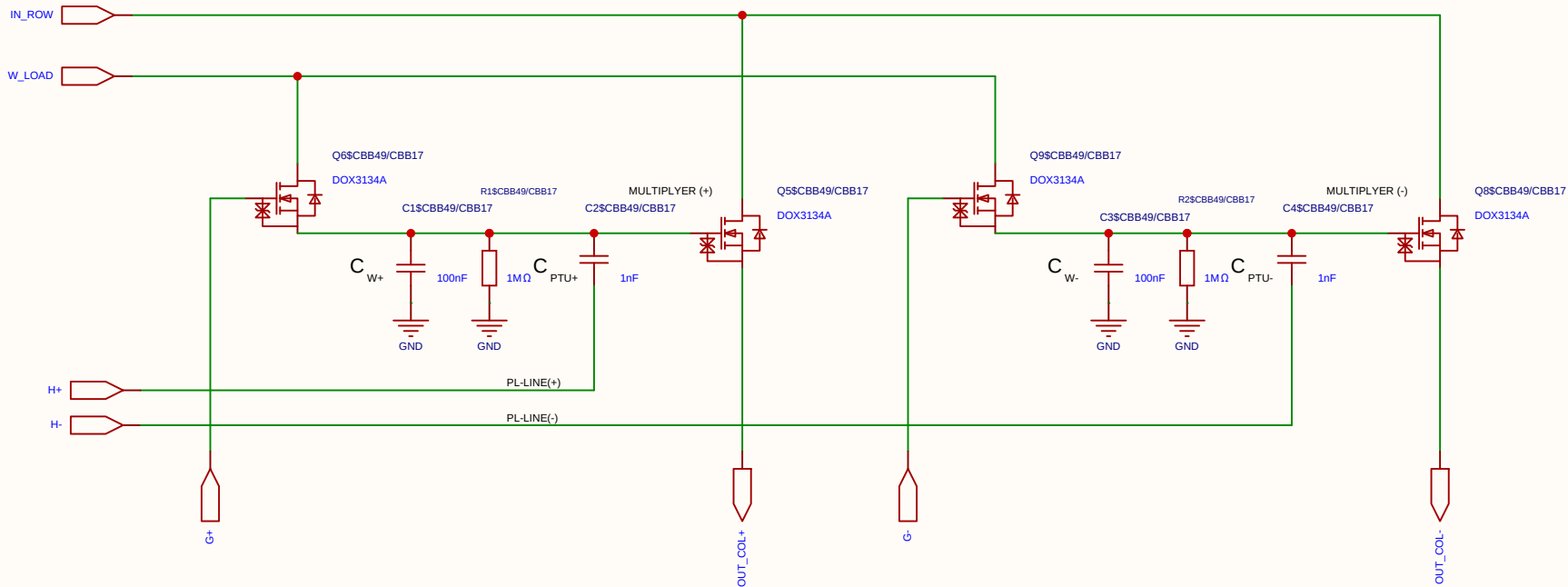
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

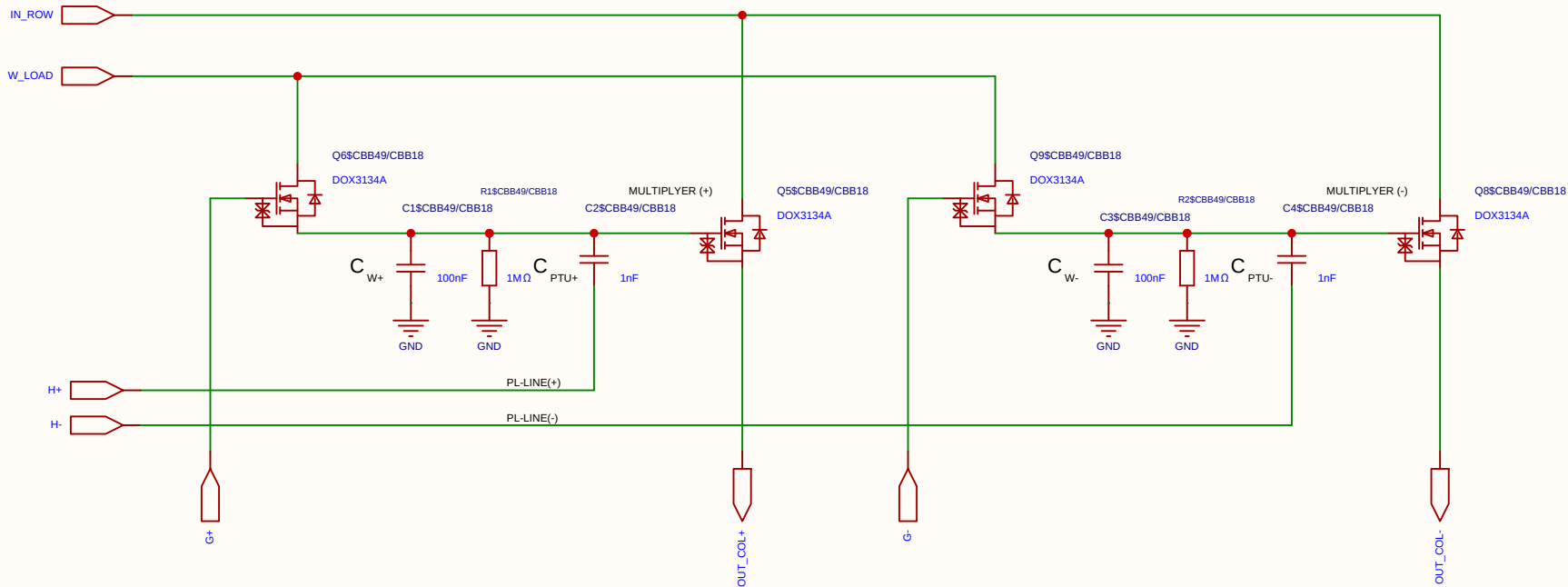
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

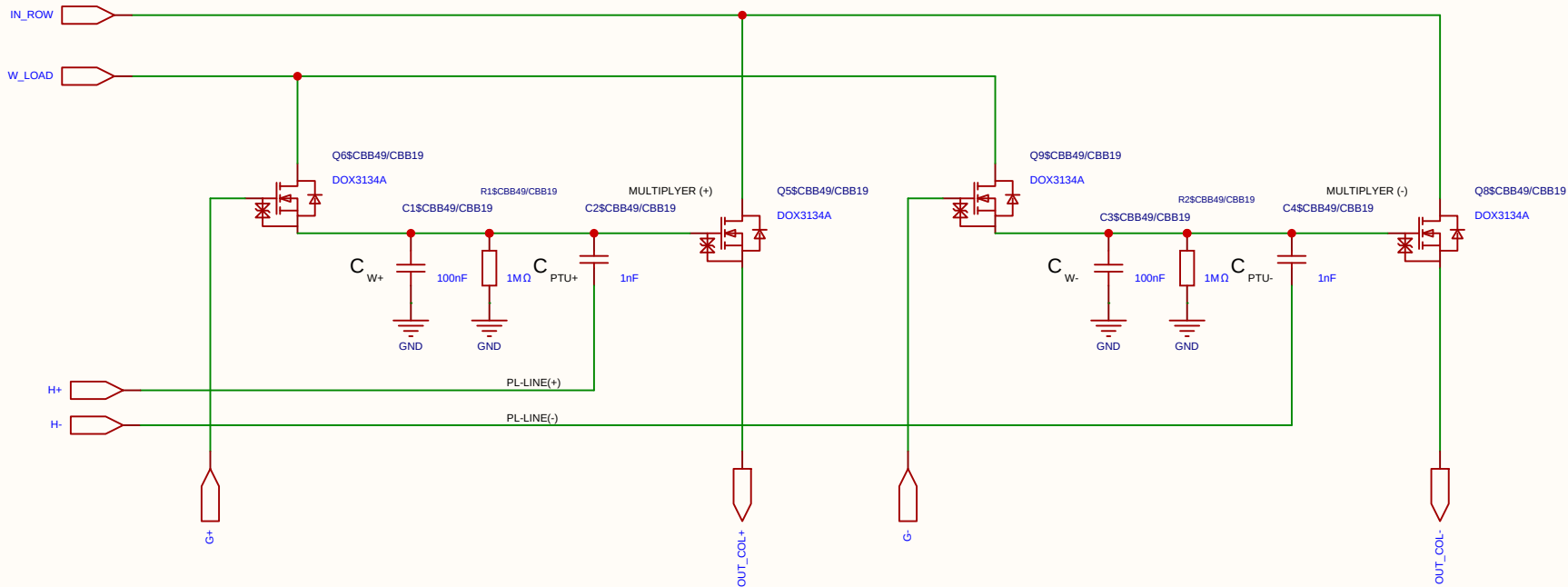
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

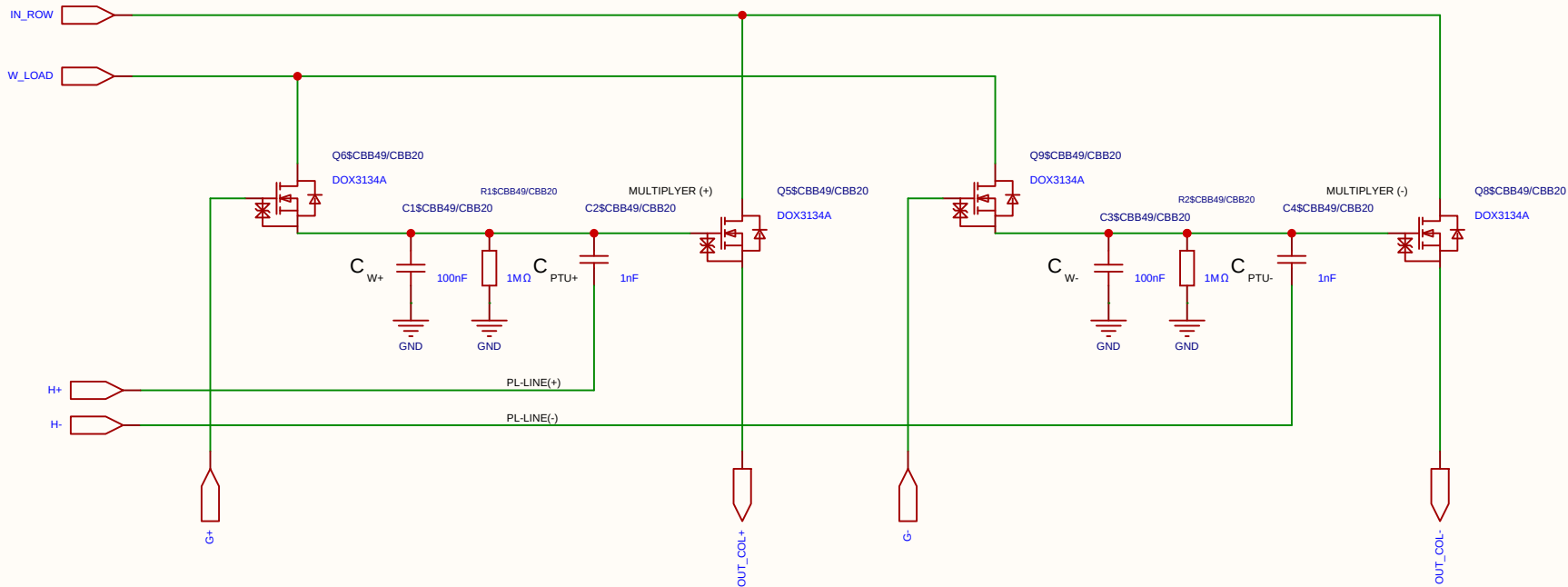
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

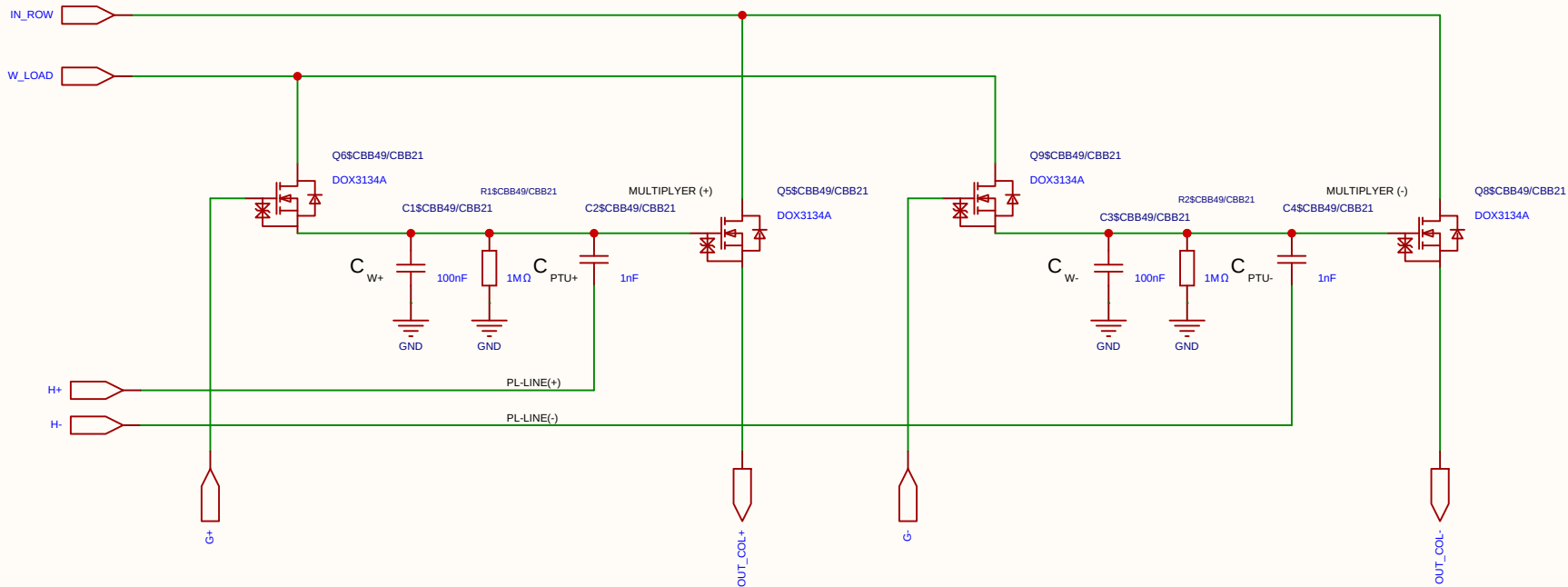
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

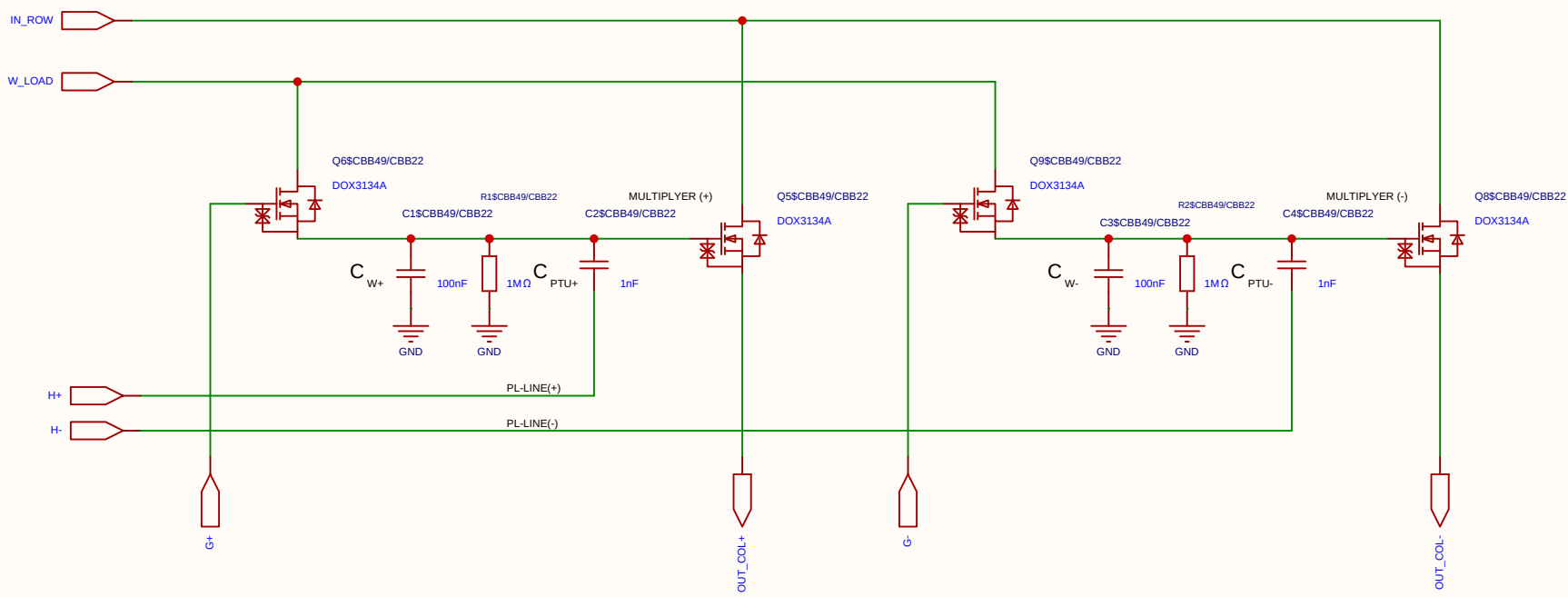
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

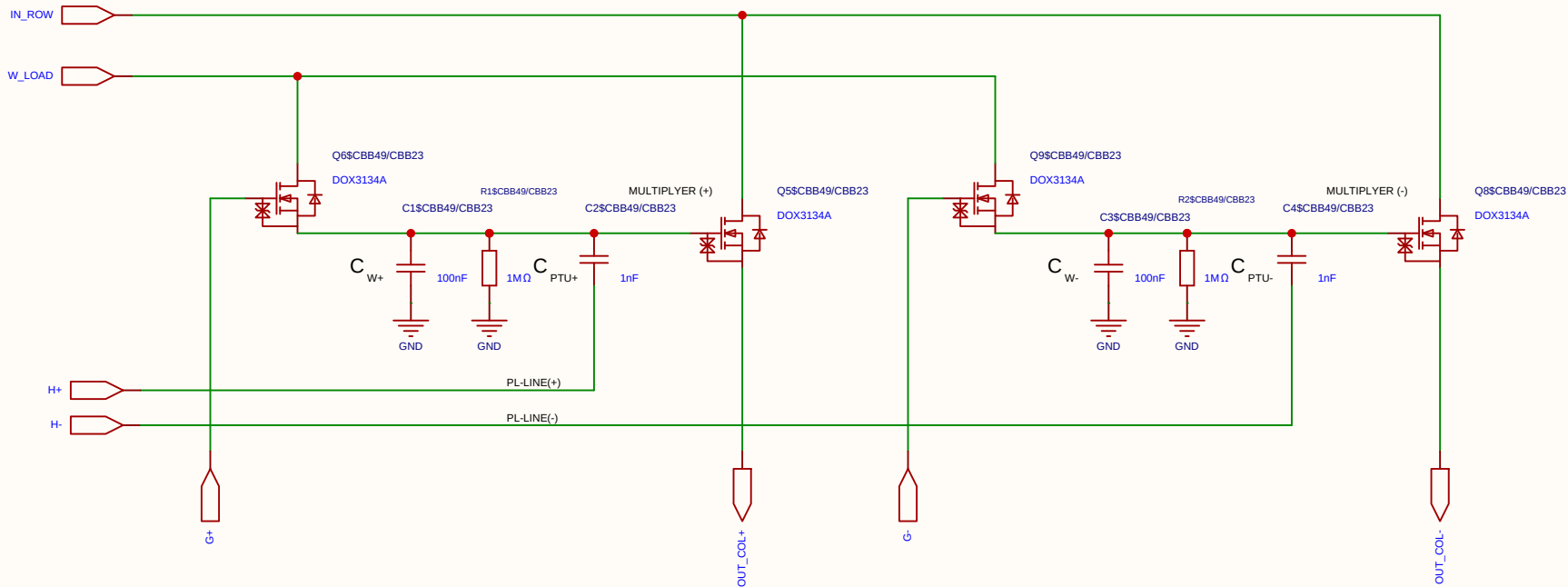
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

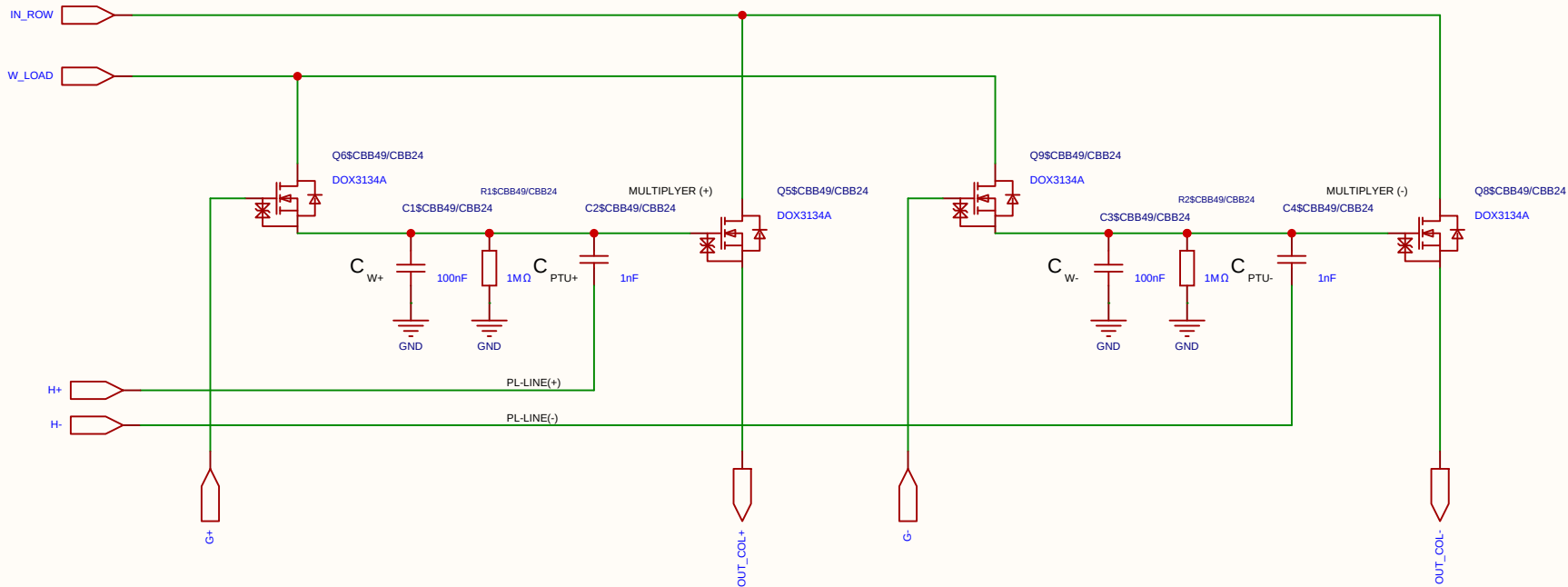
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

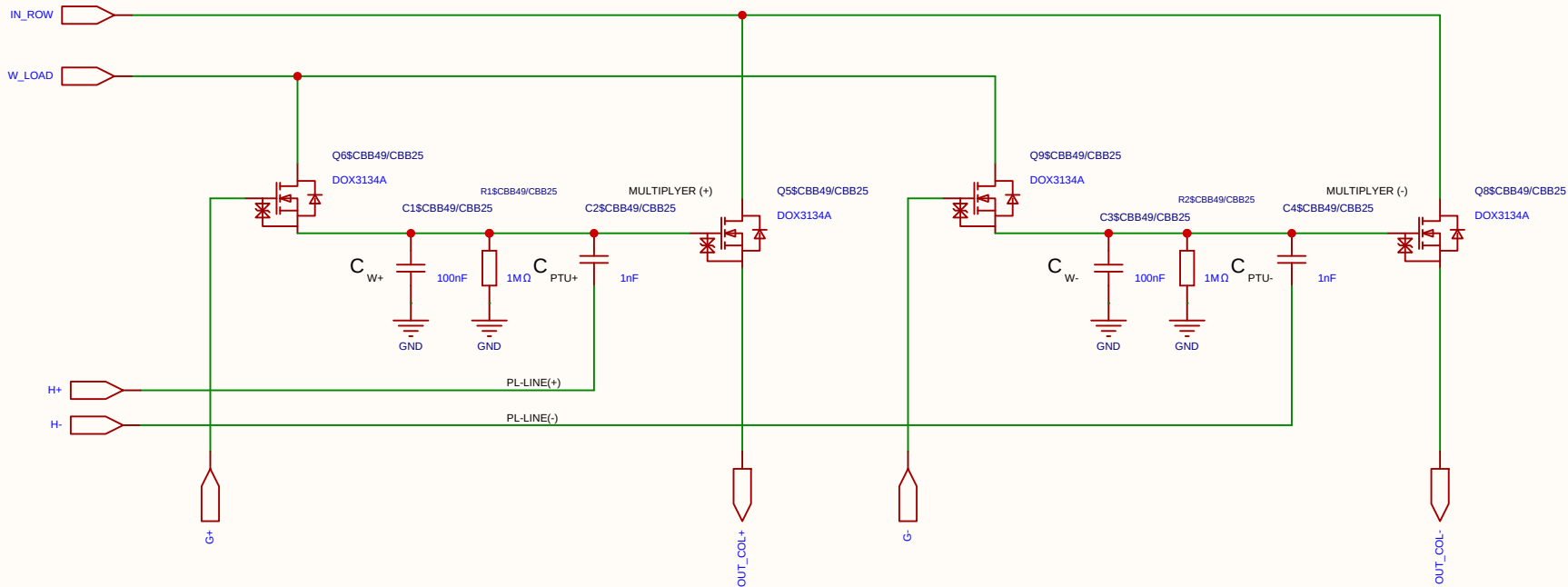
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

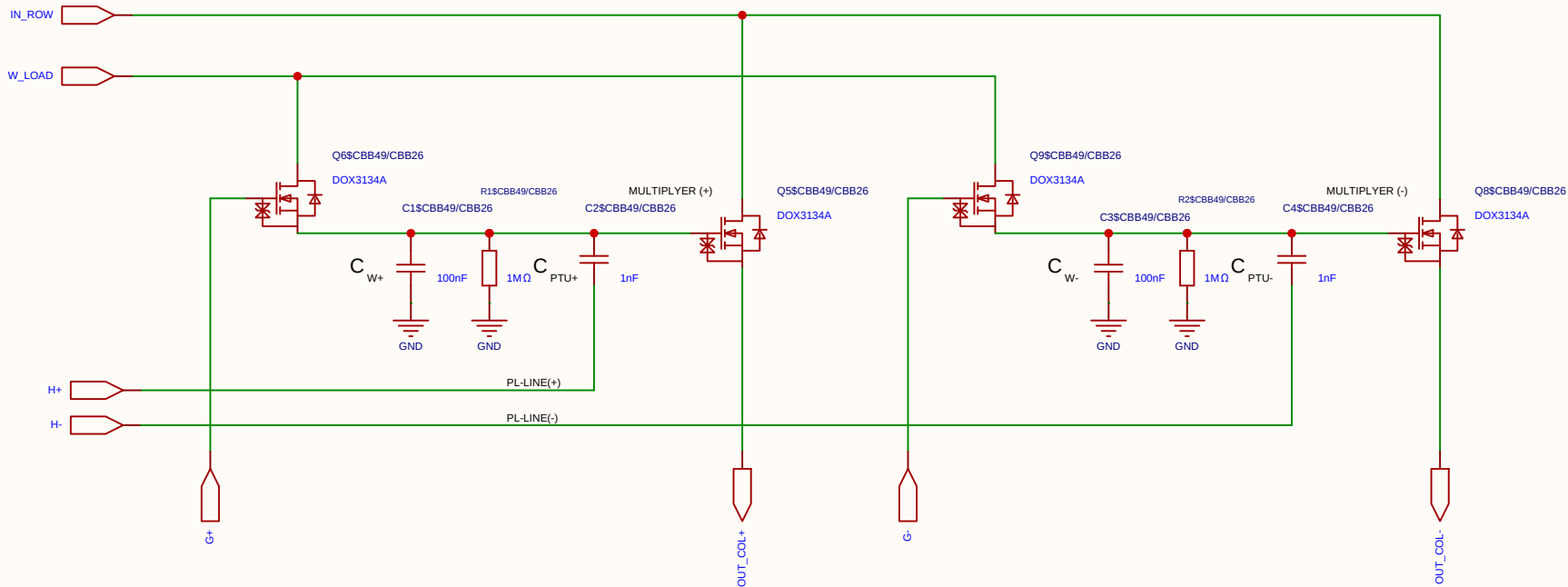
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

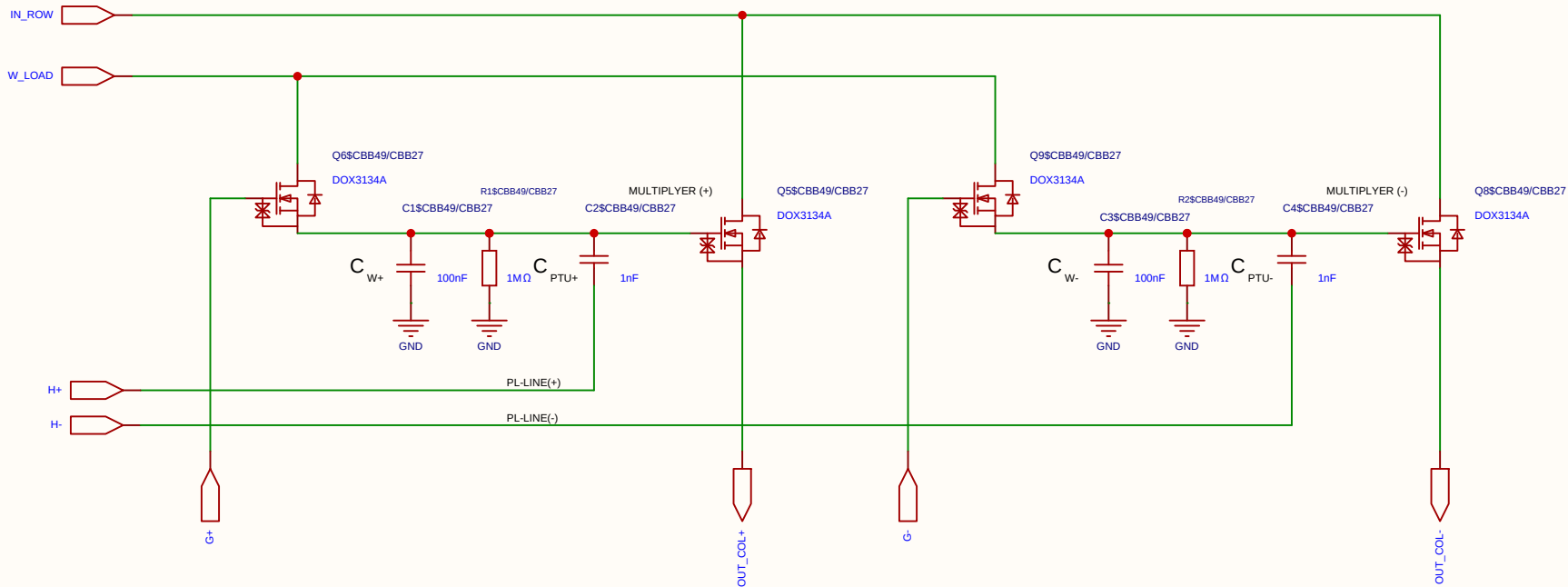
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

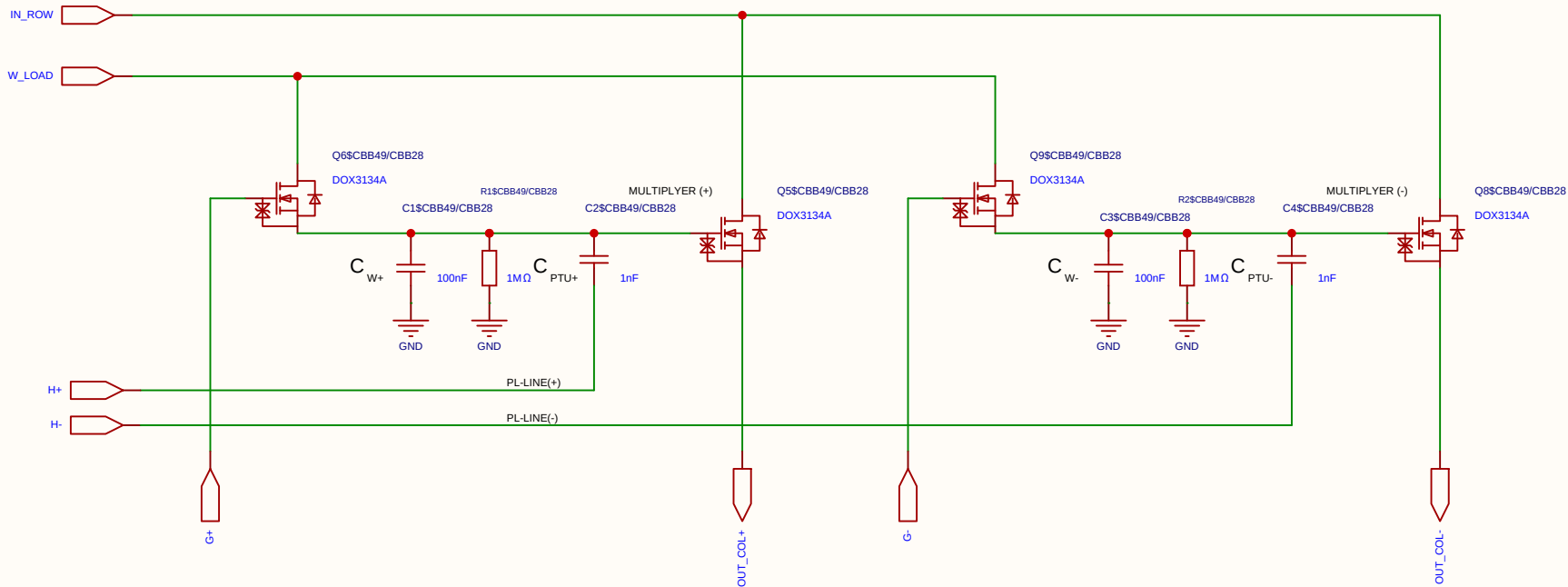
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

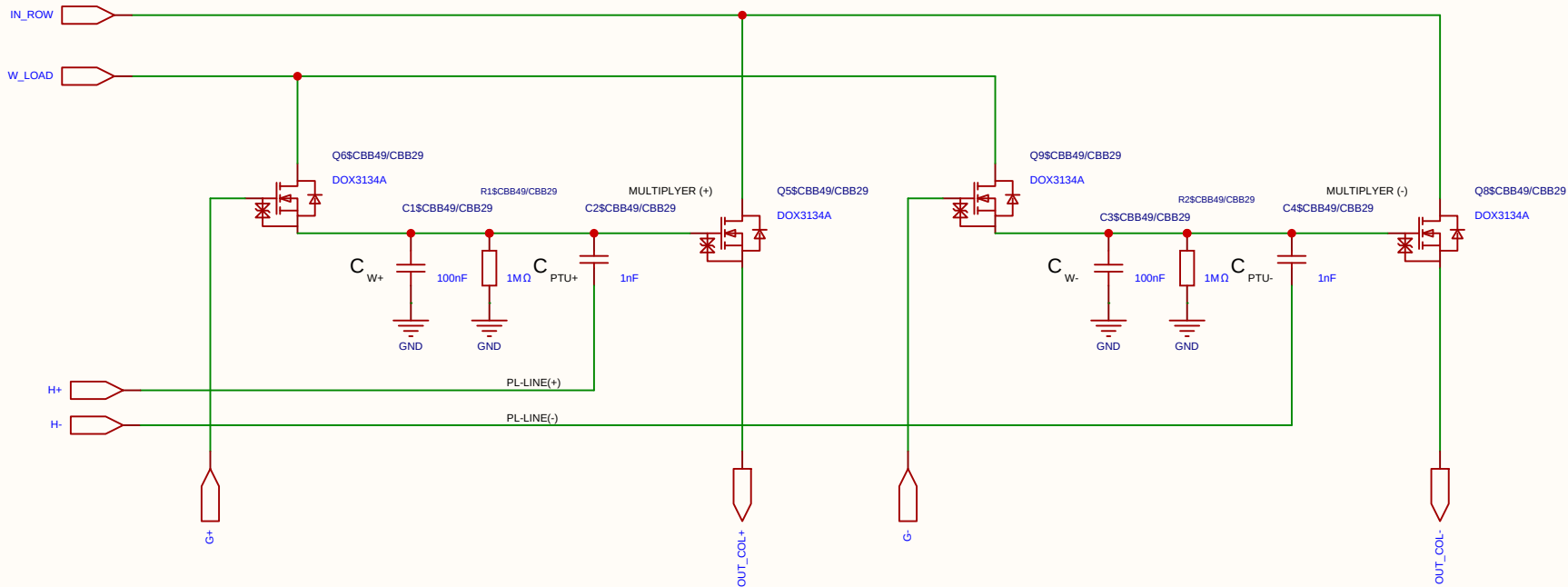
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

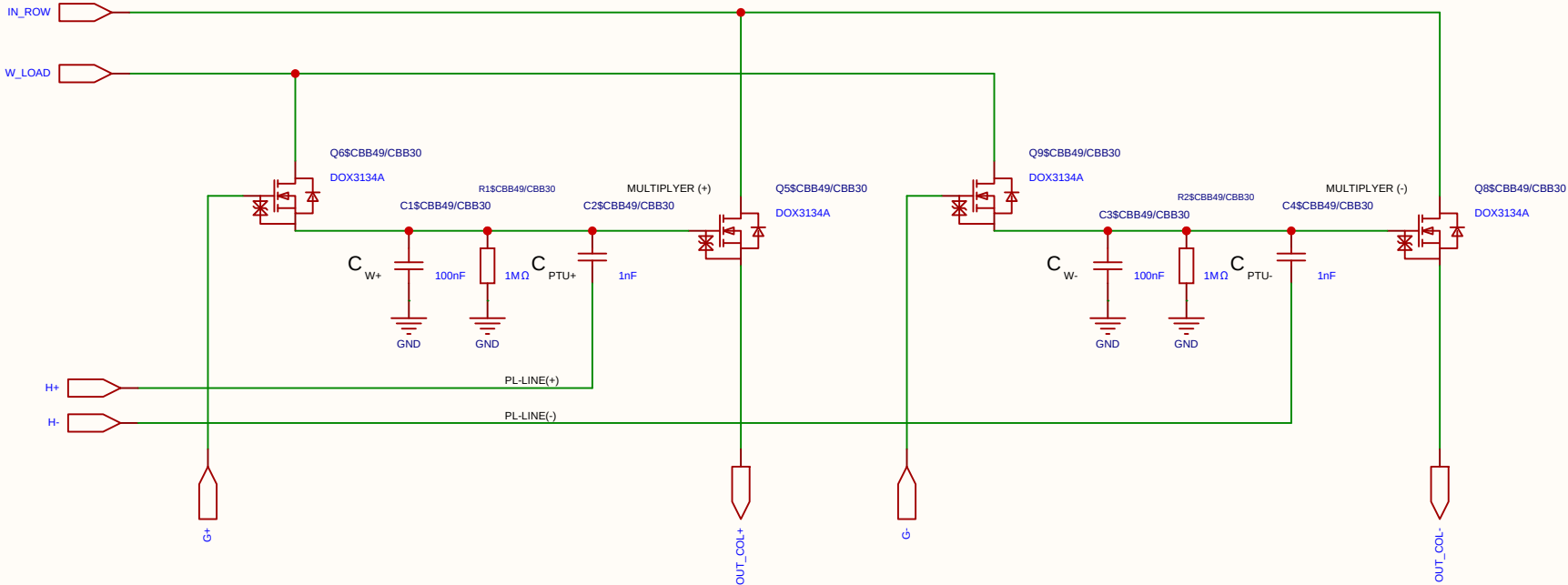
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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EasyEDA		V1.0	A4	EasyEDA.com	

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
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The schematic diagram illustrates a two-stage CMOS differential amplifier. The circuit is powered by **IN_ROW** and **W_LOAD**. It features two input stages, each with a PMOS load (Q6, Q9) and an NMOS input pair (Q4, Q5). The inputs are **H+** and **H-**. The outputs are **OUT_COL+** and **OUT_COL-**. The circuit includes various capacitors (C1, C2, C3, C4, C_{W+}, C_{W-}), resistors (R1, R2), and a multiplier block (MULTIPLYER (+) and MULTIPLYER (-)).

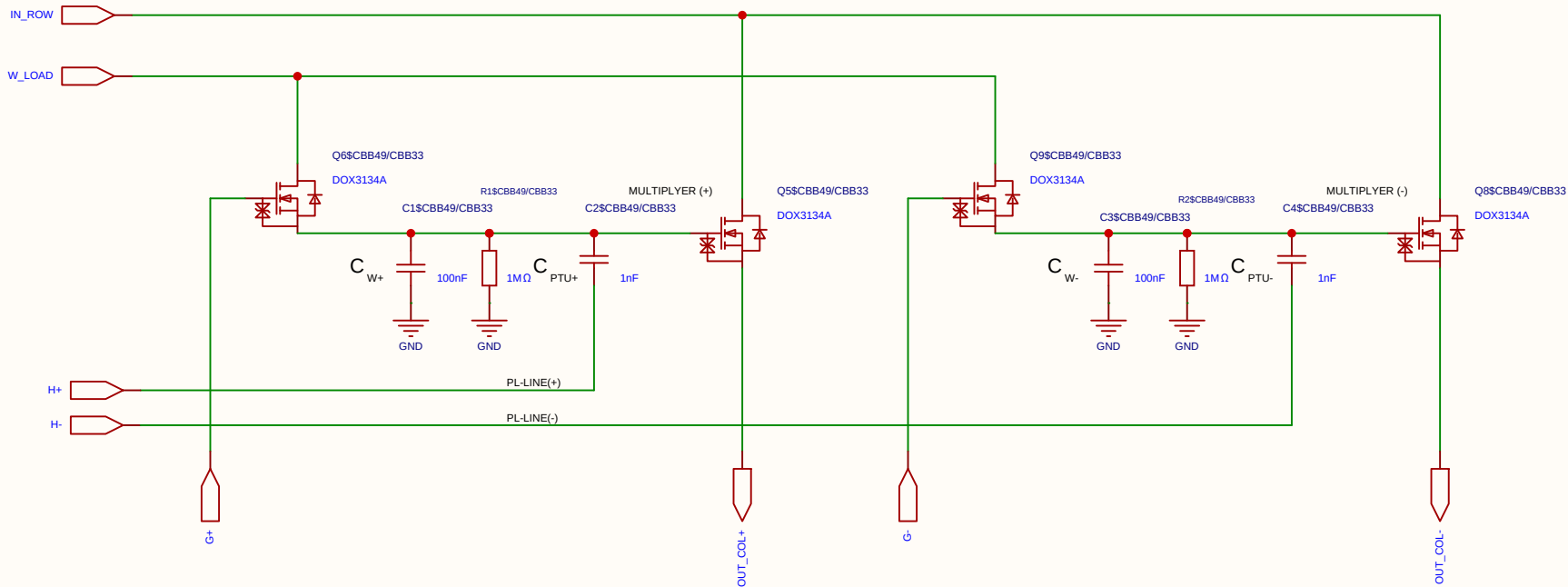
Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

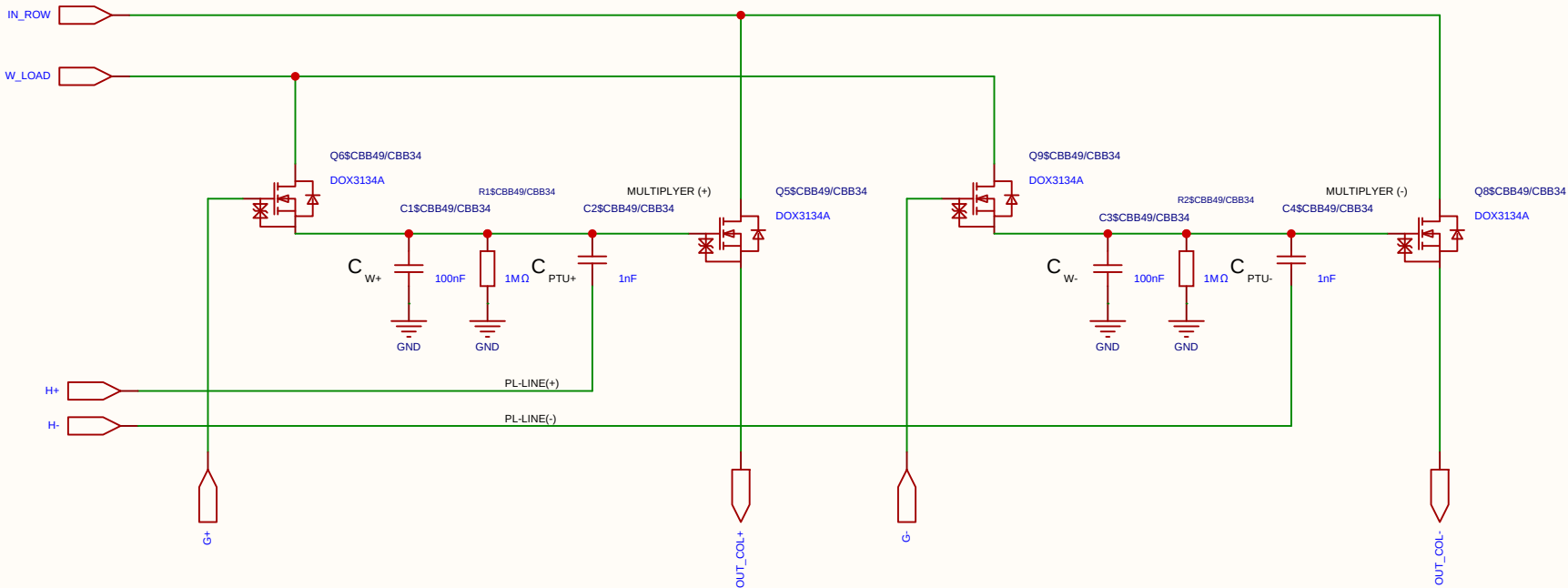
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

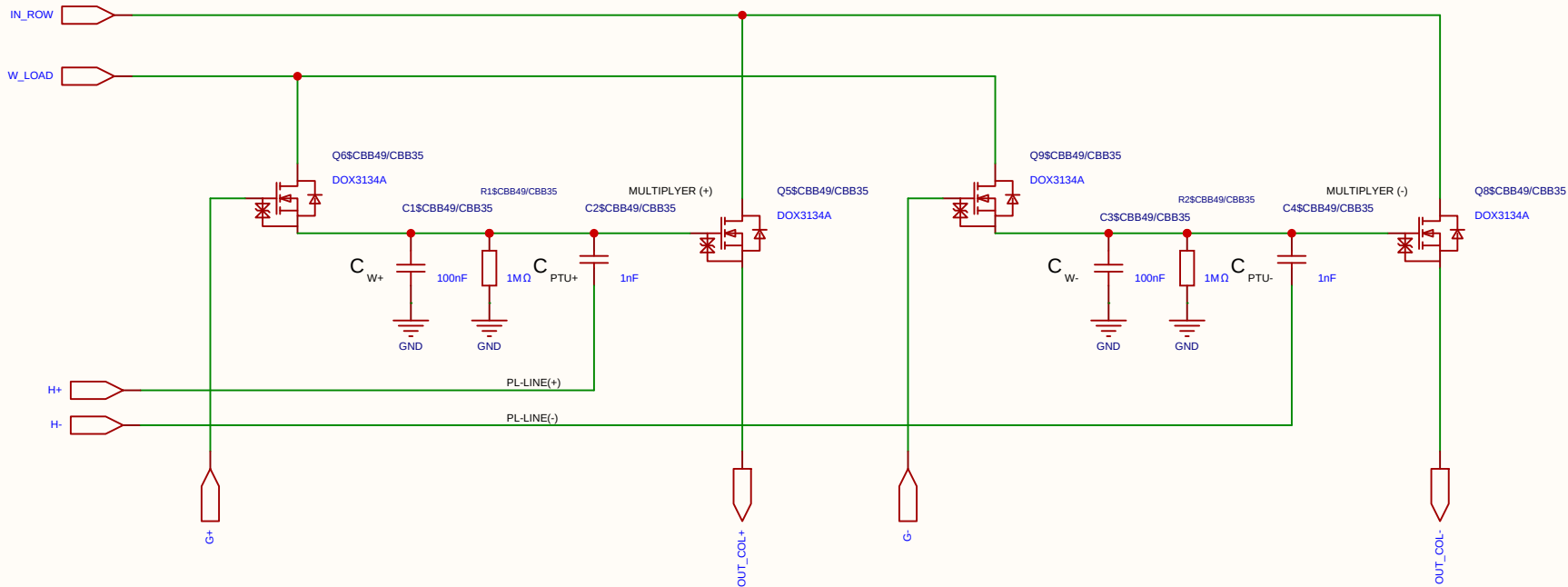
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

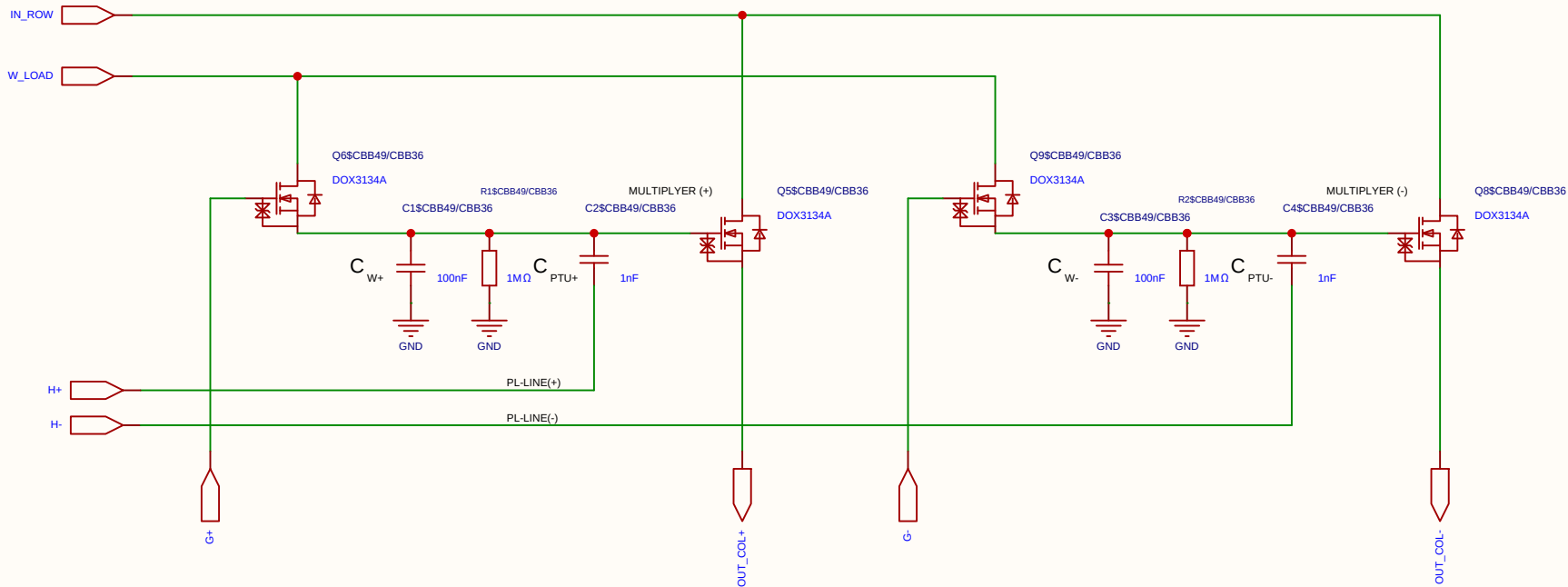
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

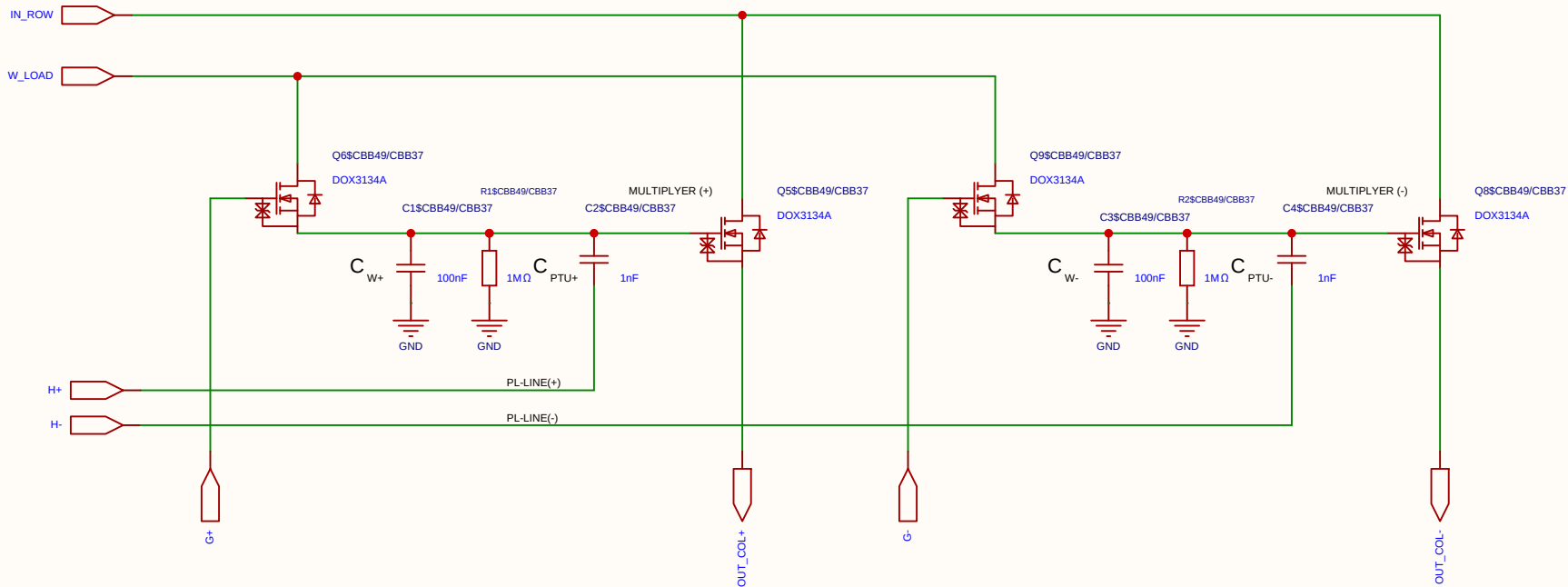
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

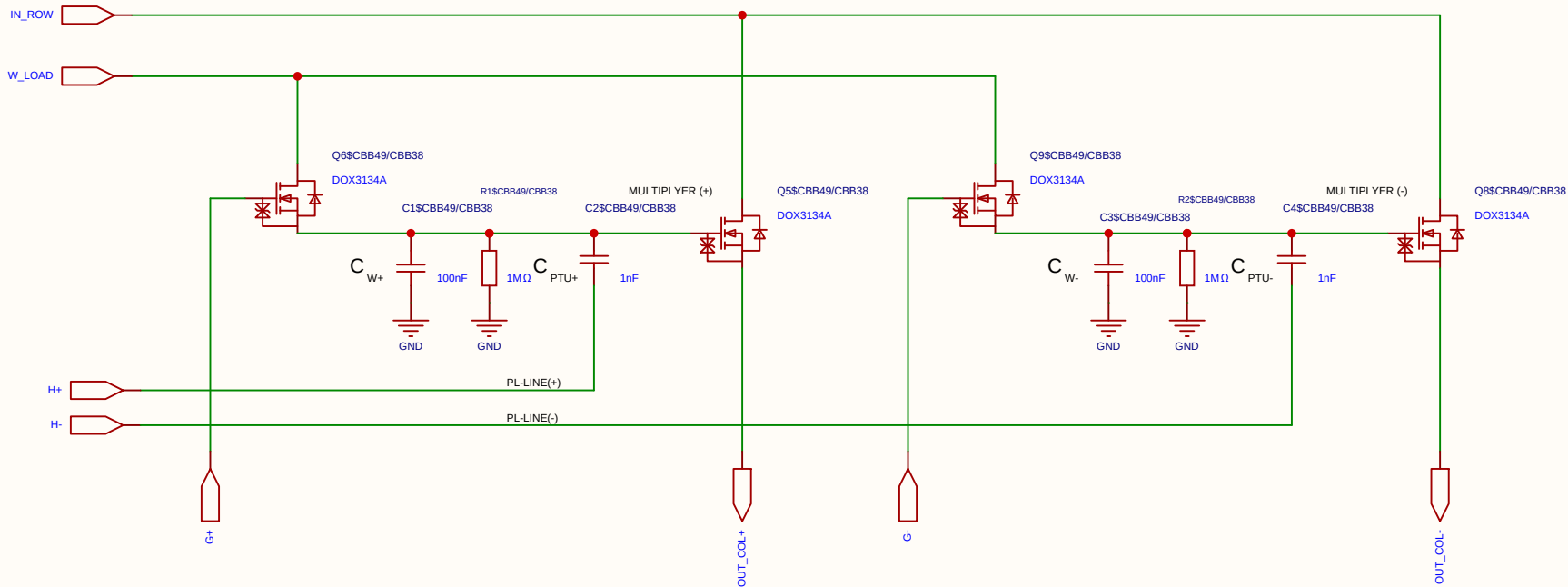
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

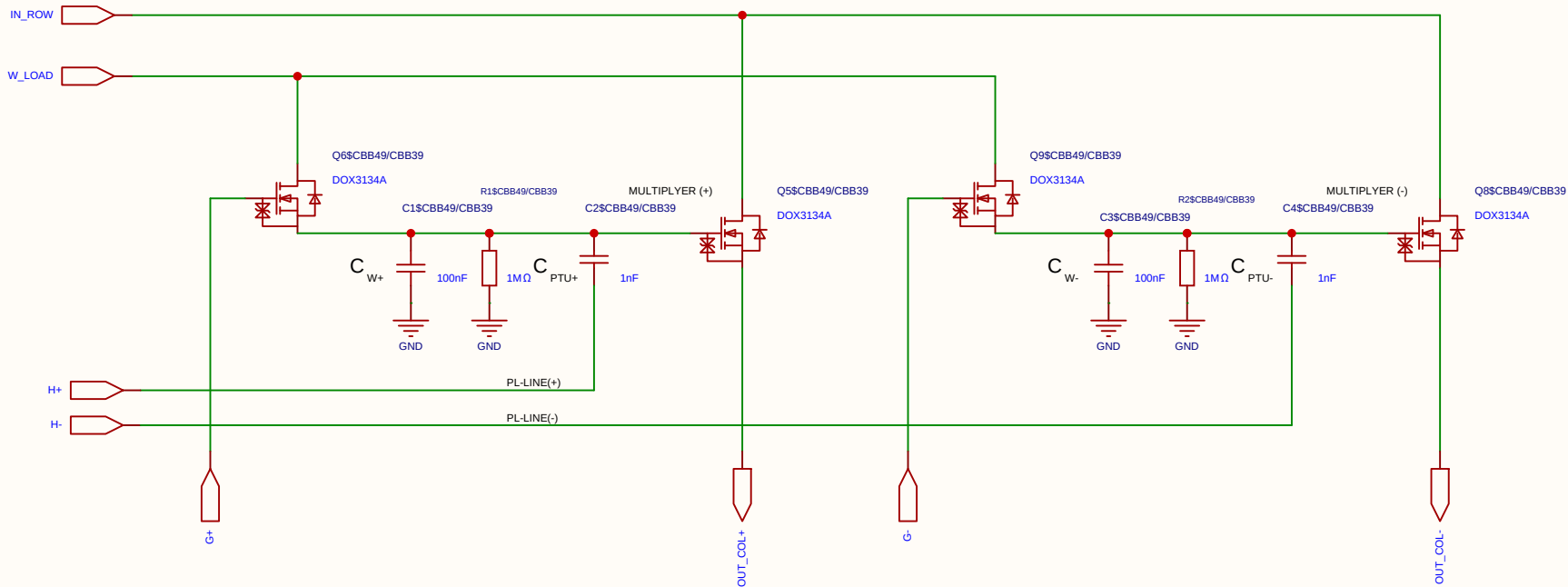
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

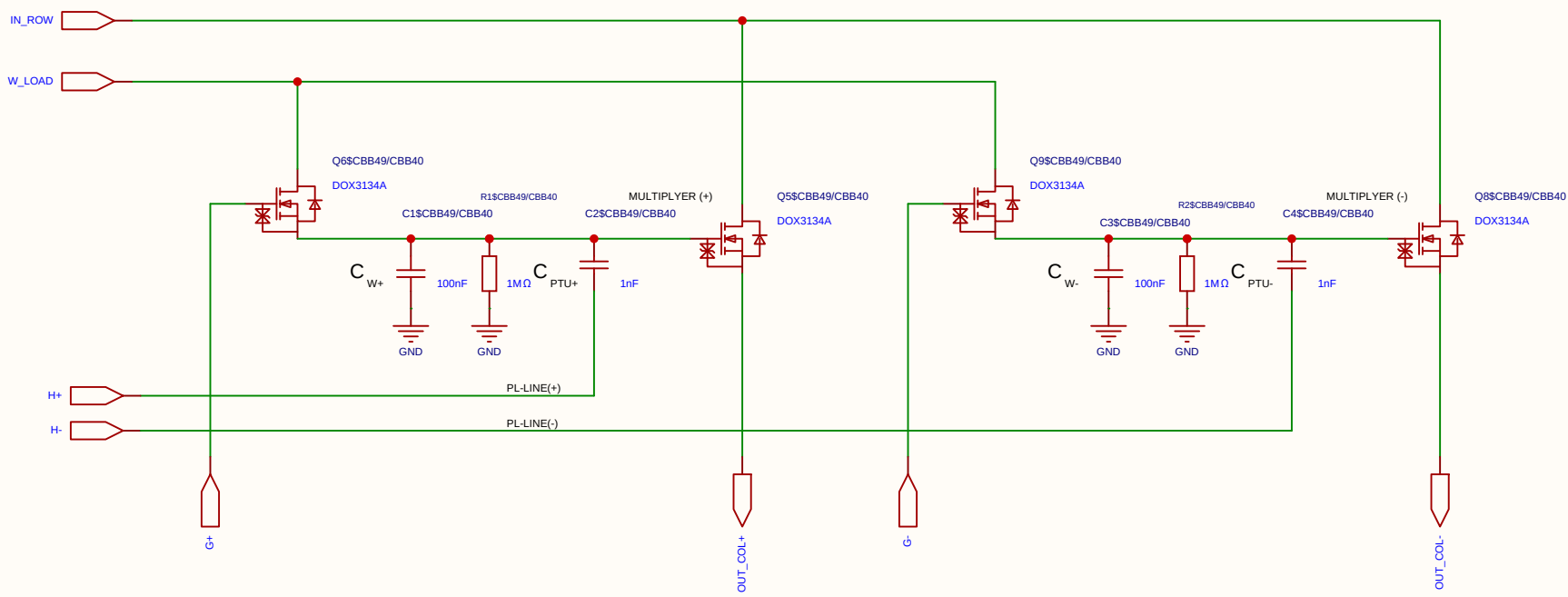
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
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		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

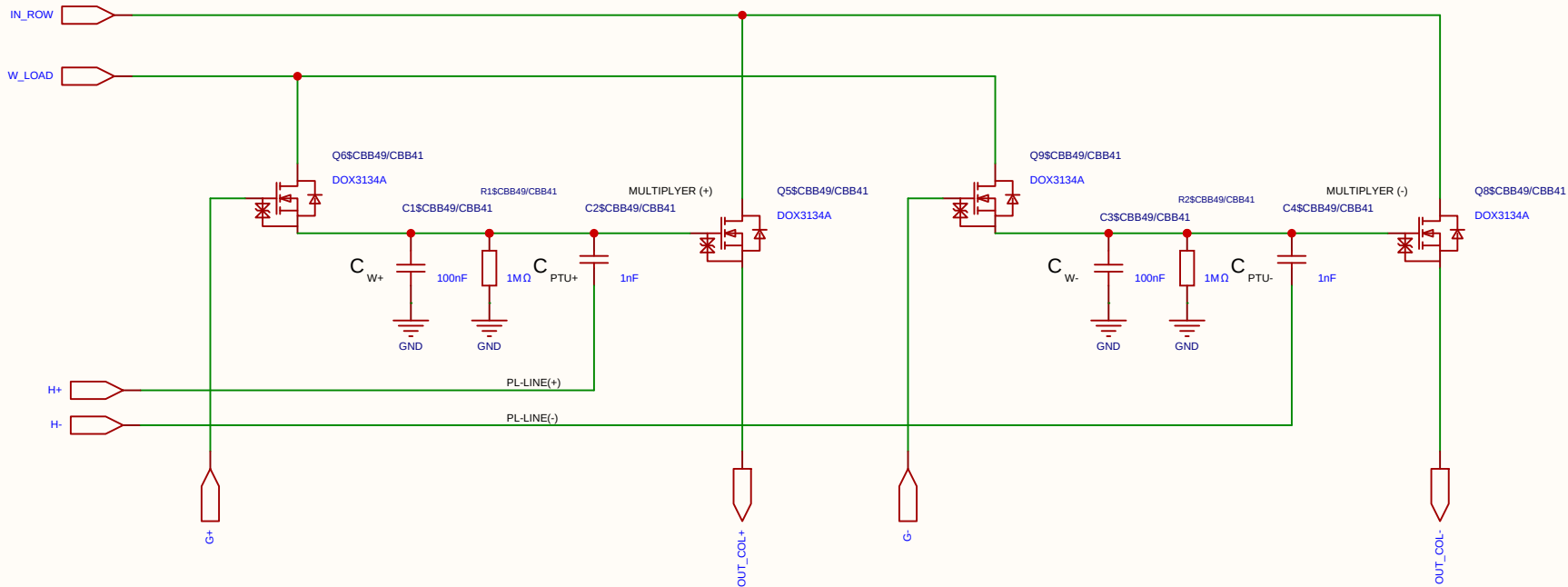
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

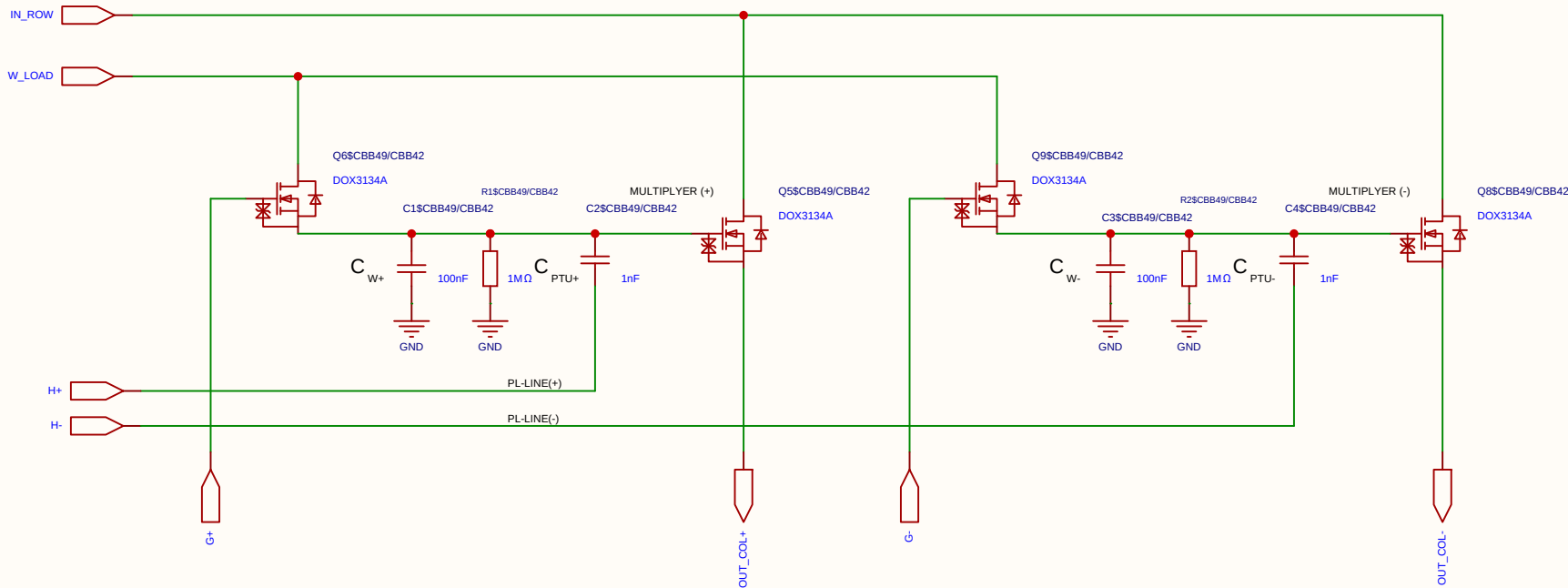
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-07
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

The schematic diagram illustrates the internal structure of the output drivers of the 6502A microprocessor. It shows two main output paths: **OUT_COL+** and **OUT_COL-**. Each path is driven by a PMOS transistor (Q6\$CBB49/CBB43 and Q8\$CBB49/CBB43) and an NMOS transistor (Q5\$CBB49/CBB43 and Q9\$CBB49/CBB43). The PMOS transistors are connected to the **V+** supply, and the NMOS transistors are connected to ground. The gates of the PMOS transistors are connected to the **PL-LINE(+)** signal, and the gates of the NMOS transistors are connected to the **PL-LINE(-)** signal. The drains of the NMOS transistors are connected to the **OUT_COL+** and **OUT_COL-** signals. The schematic also shows the internal compensation network, including capacitors C1\$CBB49/CBB43 (100nF), C2\$CBB49/CBB43 (1nF), C3\$CBB49/CBB43 (100nF), and C4\$CBB49/CBB43 (1nF), and resistors R1\$CBB49/CBB43 (330kΩ) and R2\$CBB49/CBB43 (330kΩ). The output signals are connected to the external load through the **OUT_COL+** and **OUT_COL-** pins.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



[illegible]

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

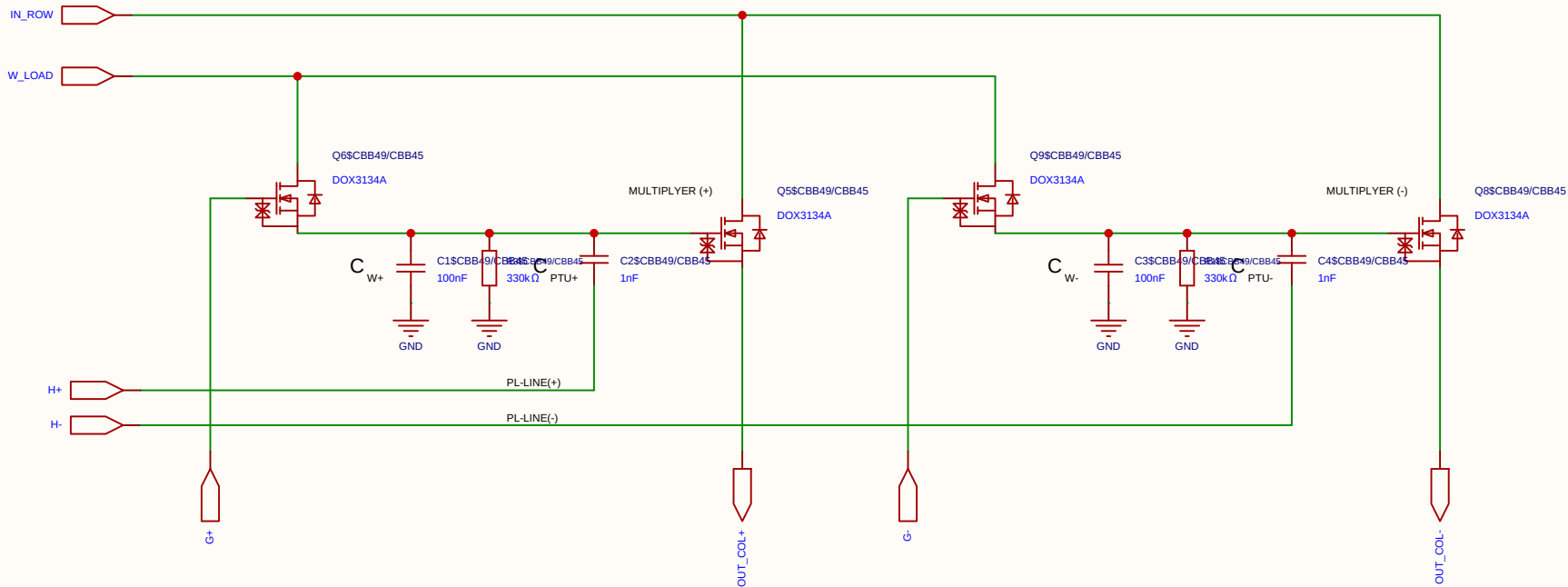
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

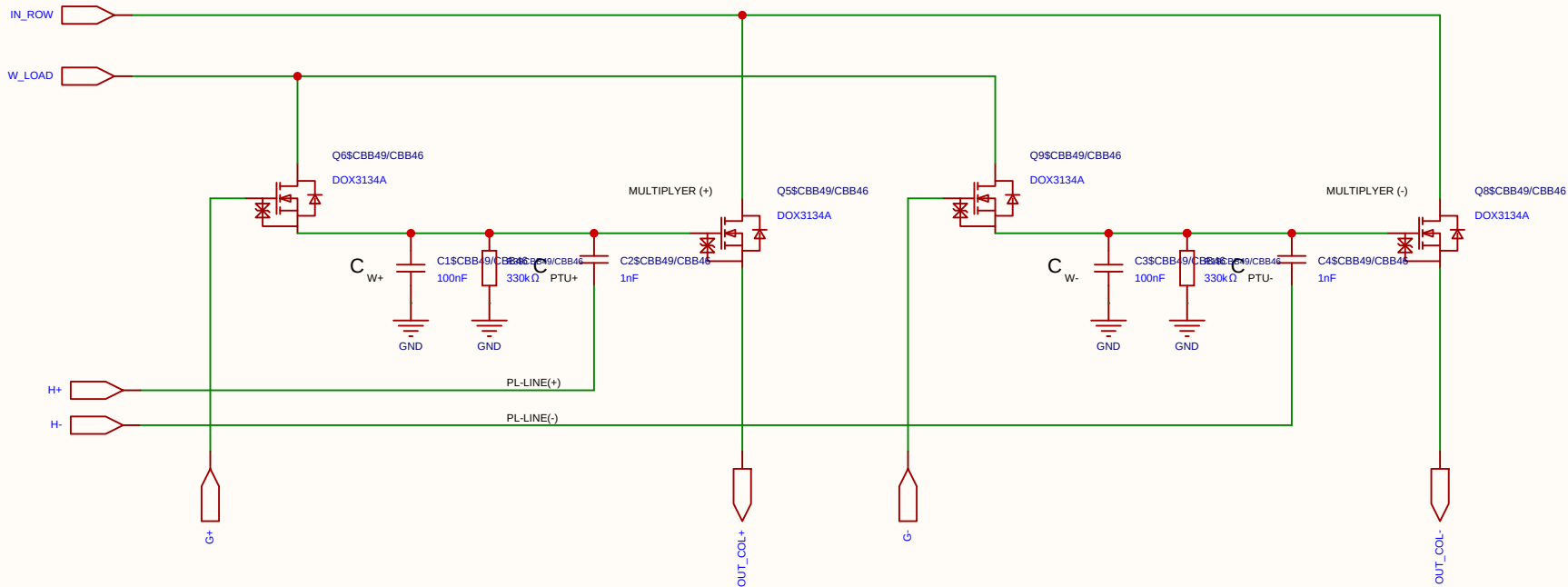
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-10
				Update at	2026-06-11
Board				Page	matrix_3_REF_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

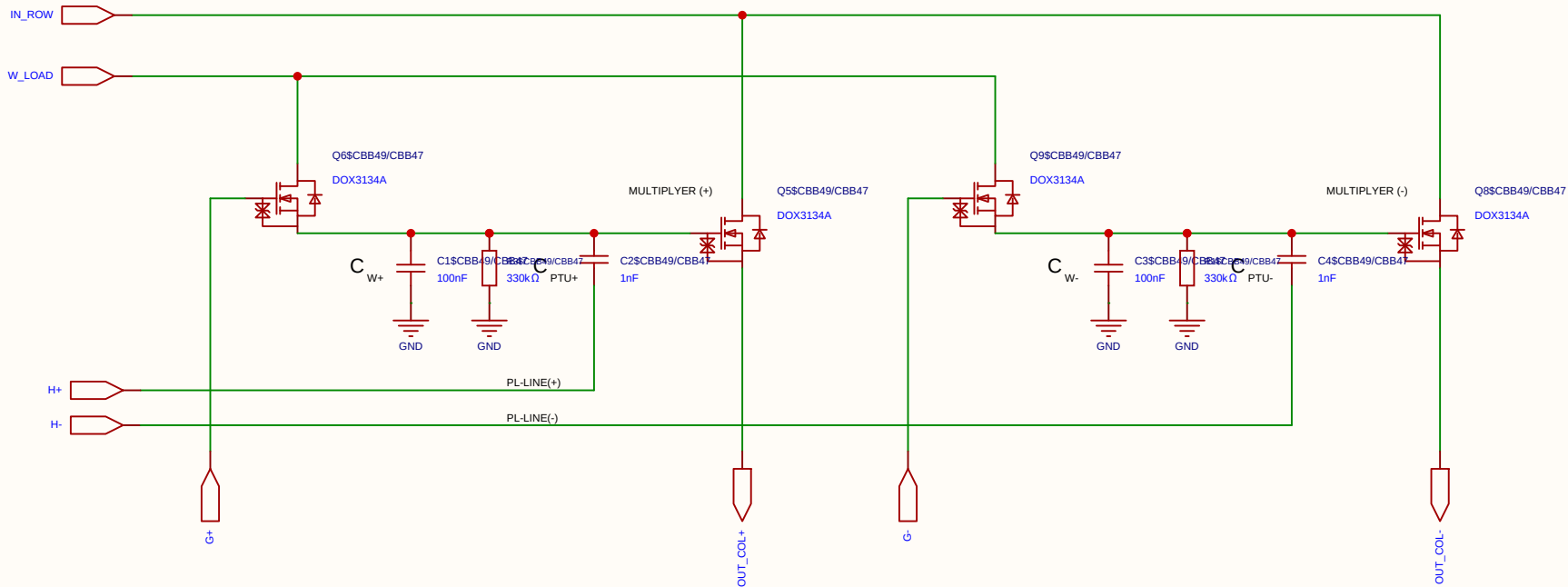
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-10
				Update at	2026-06-11
Board				Page	matrix_3_REF_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

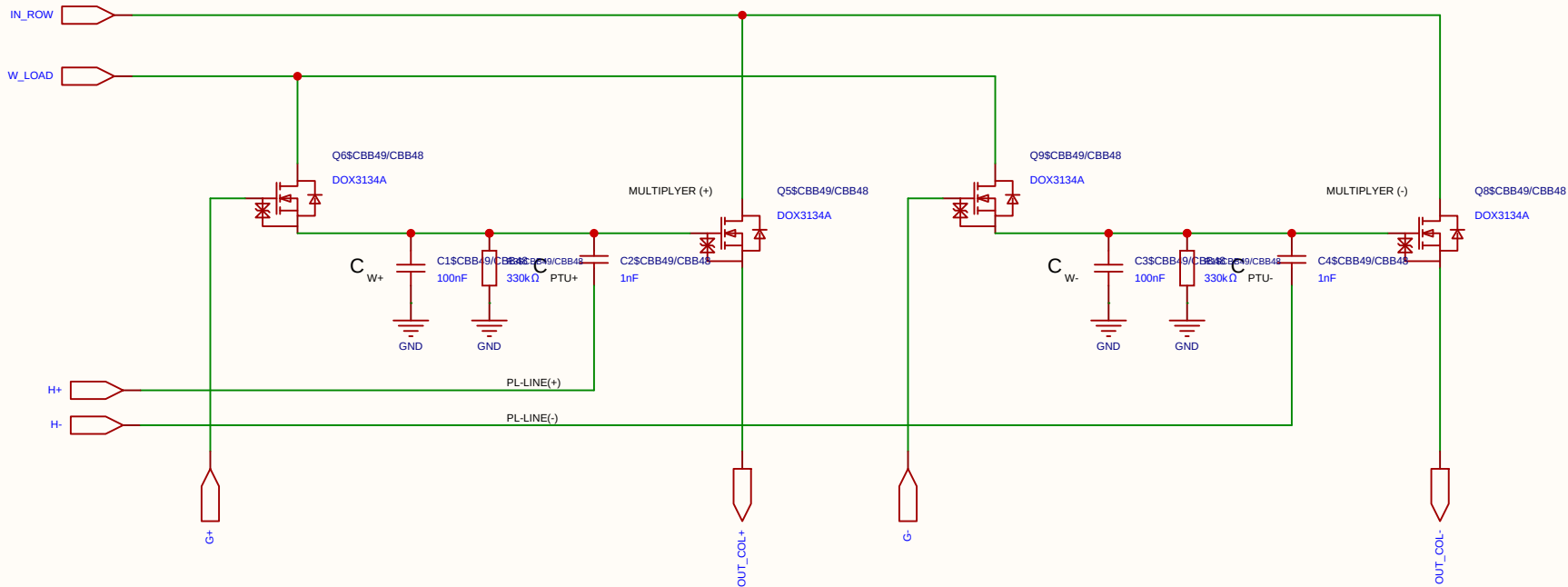
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-10
				Update at	2026-06-11
Board				Page	matrix_3_REF_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

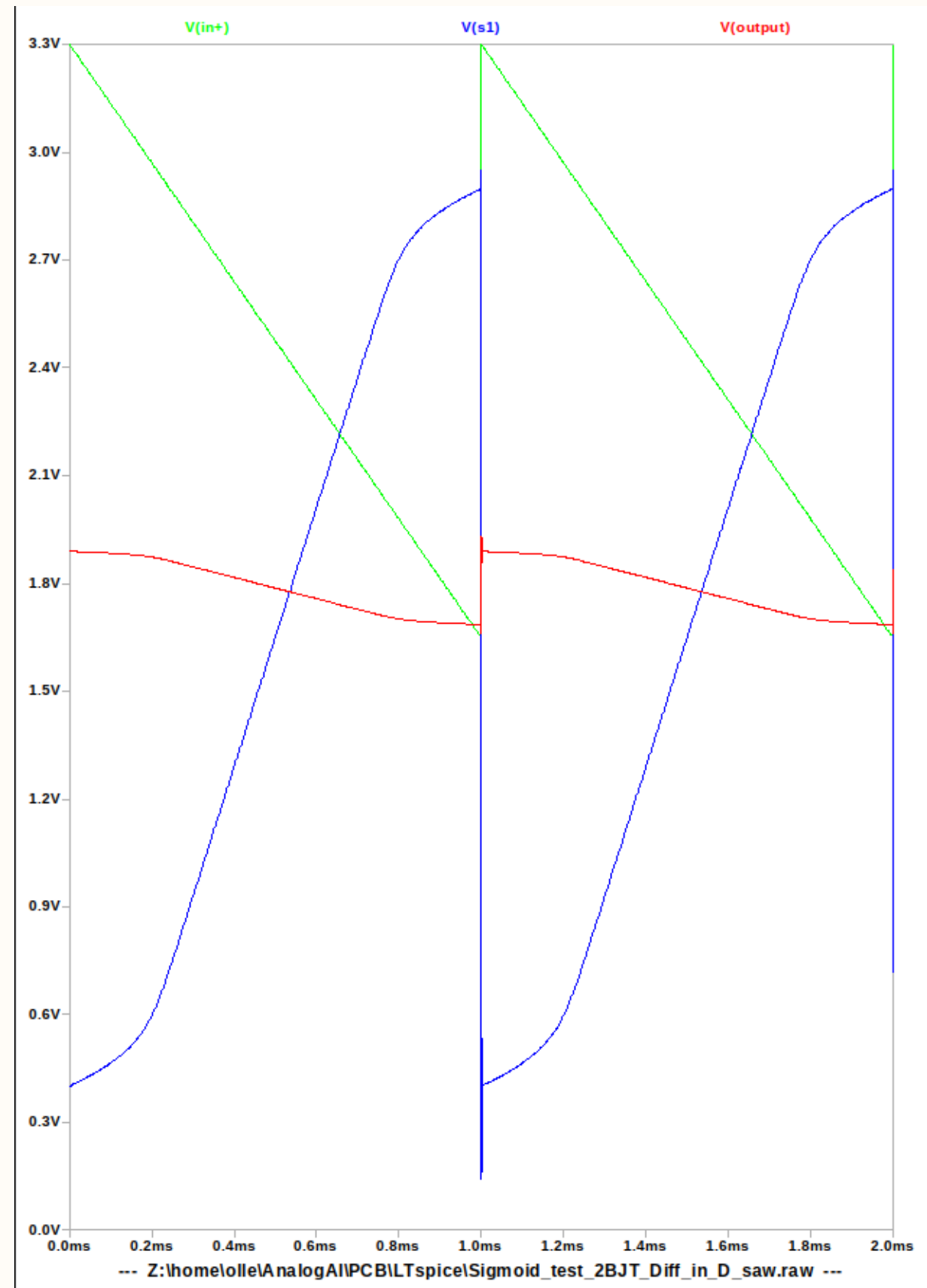
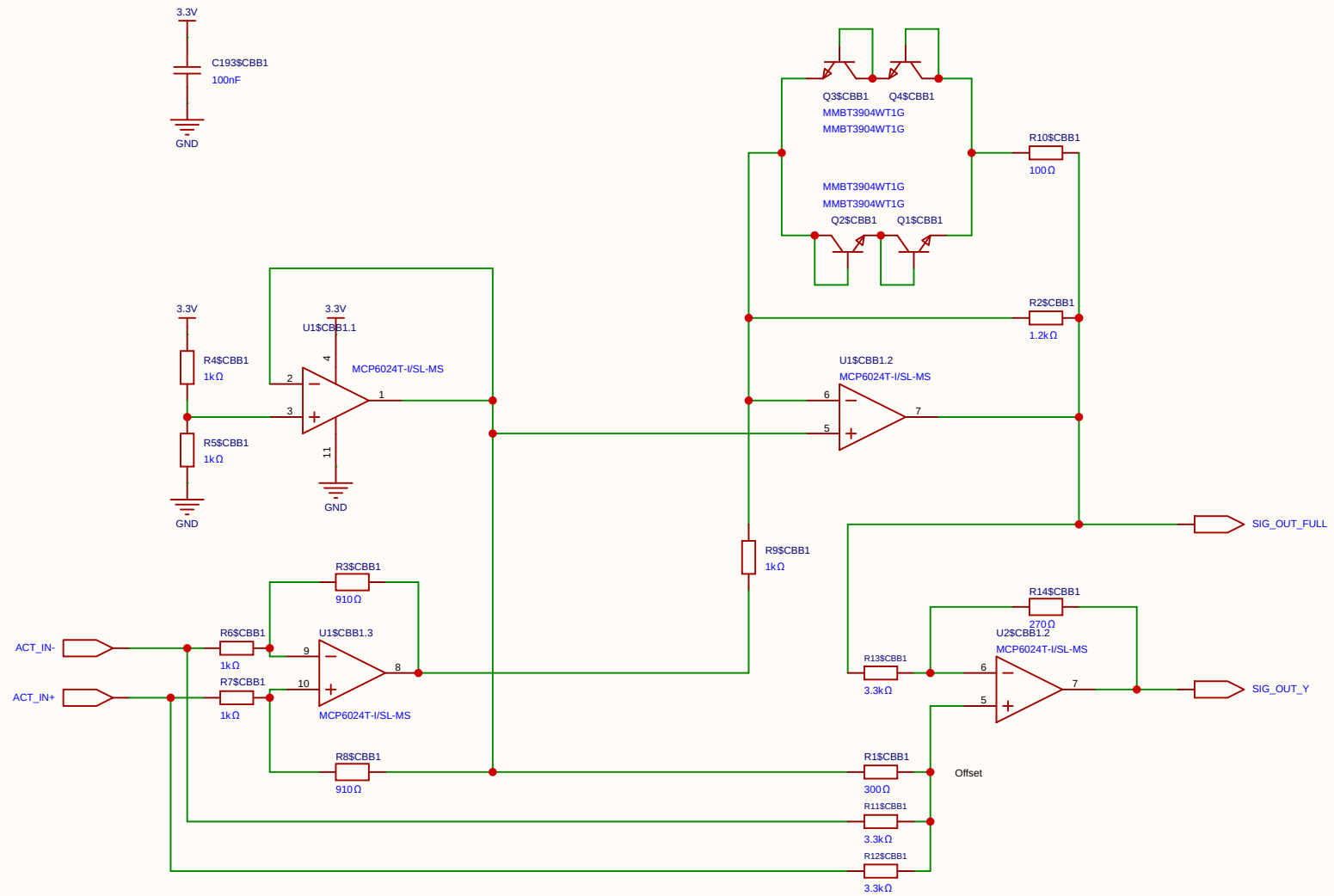
STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

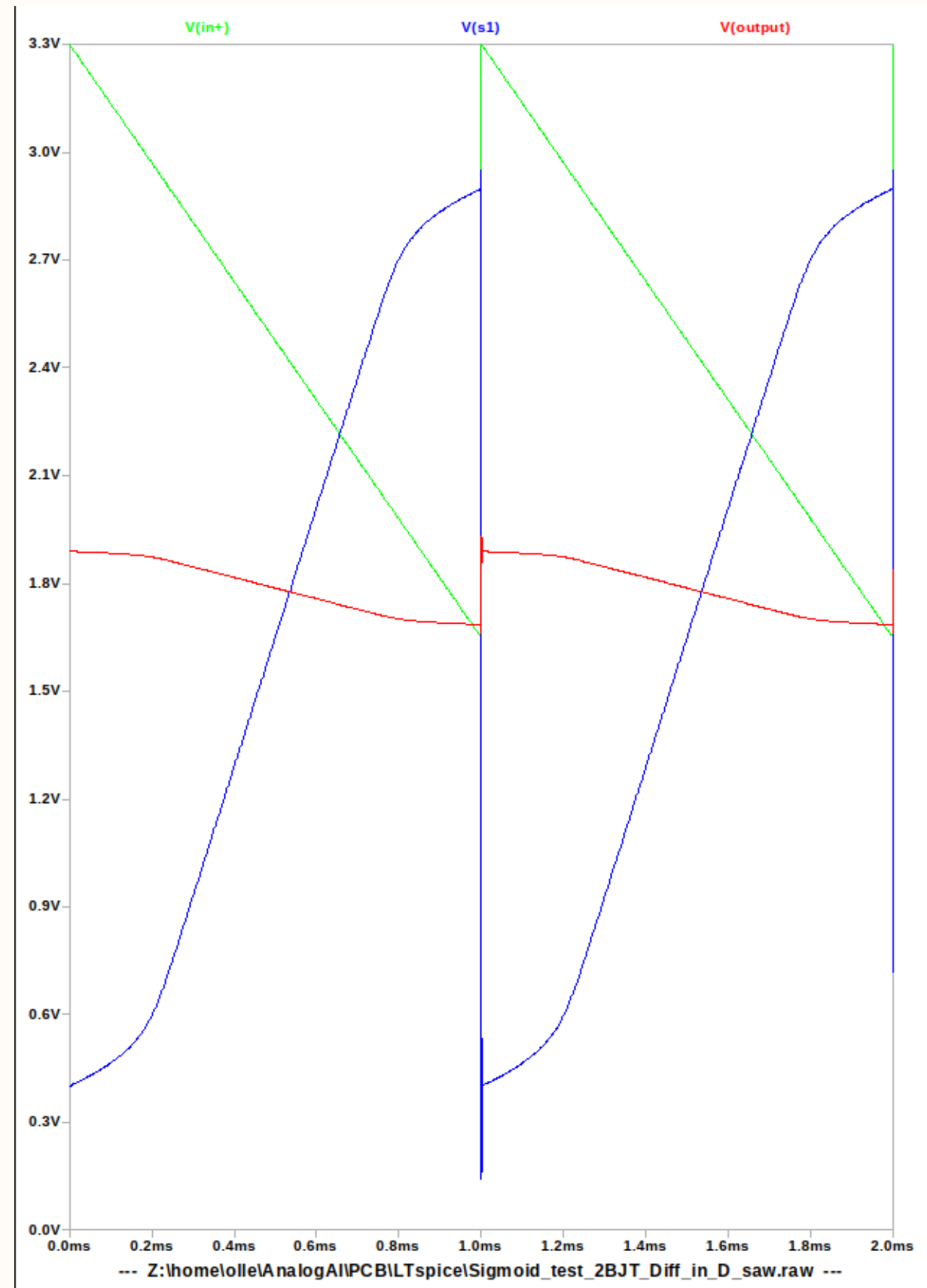
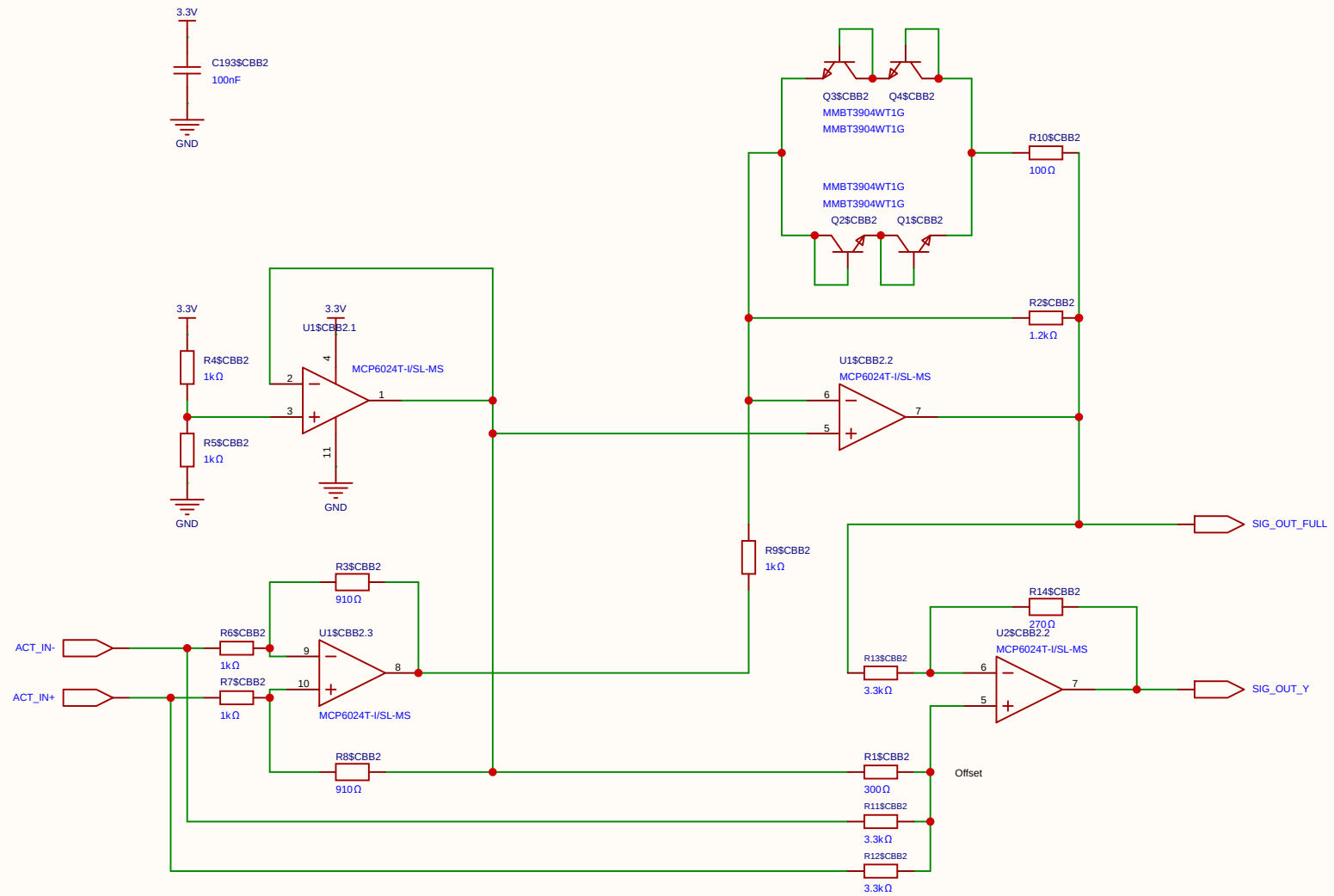
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

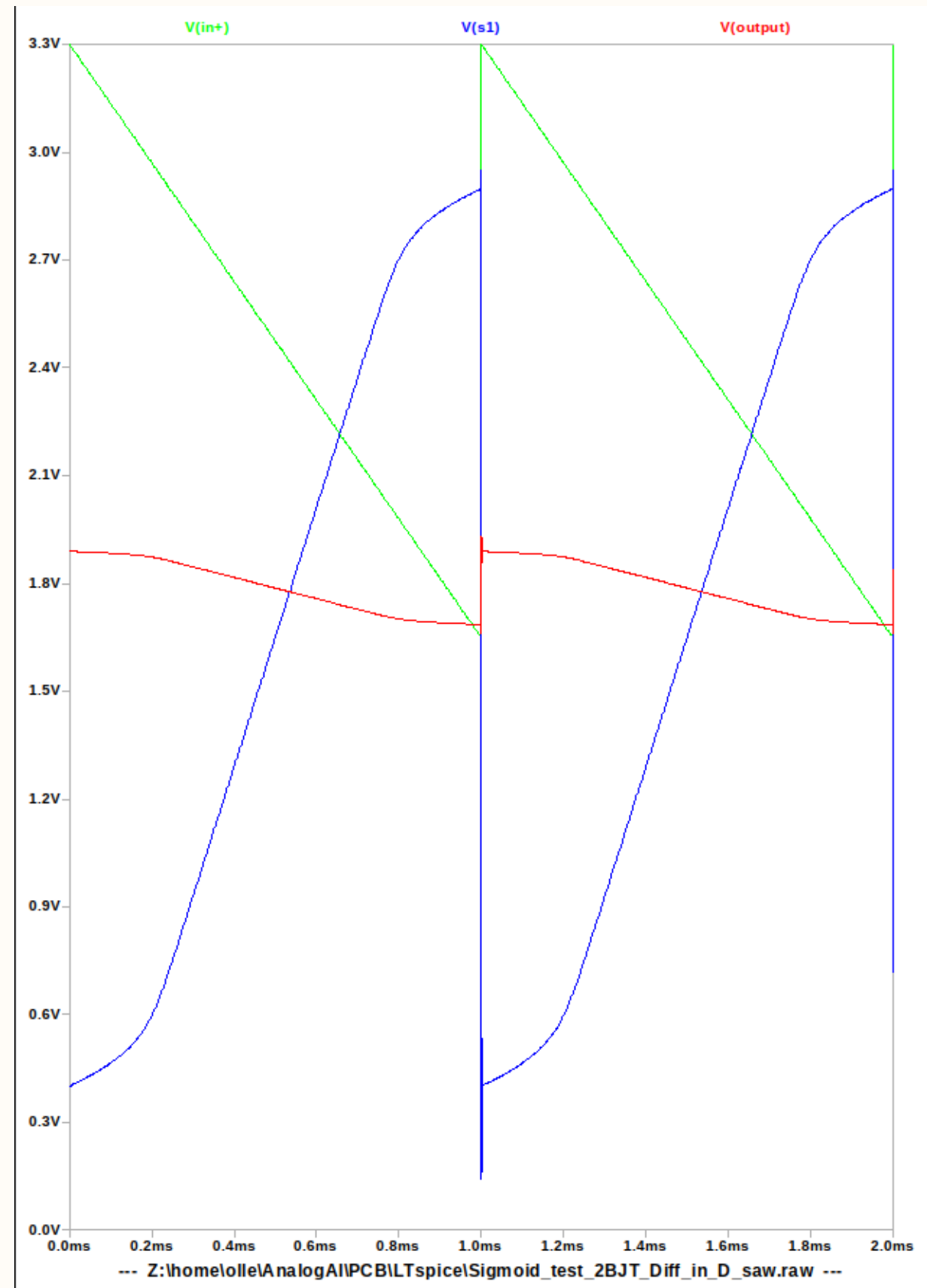
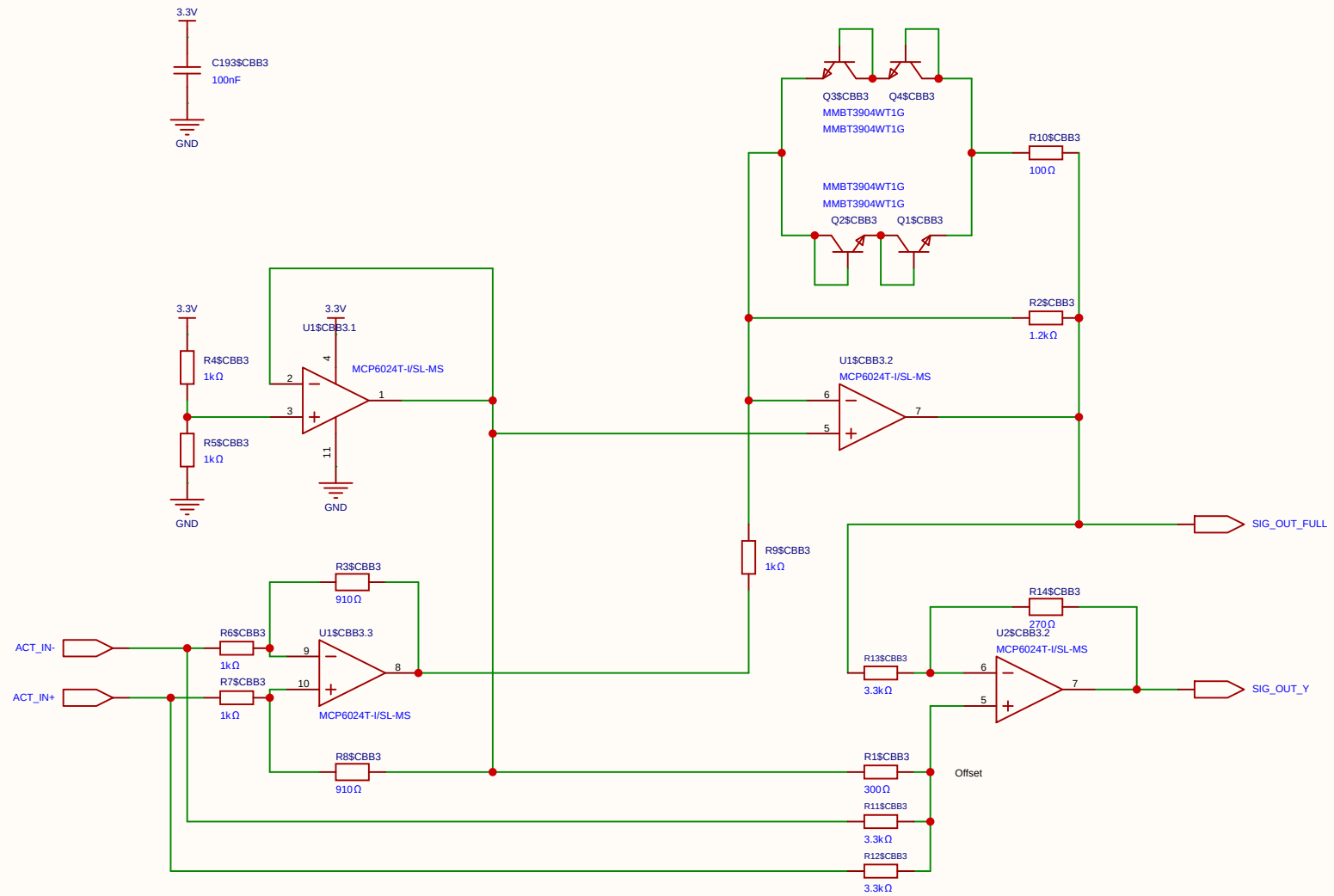
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EasyEDA		V1.0	A4	EasyEDA.com	



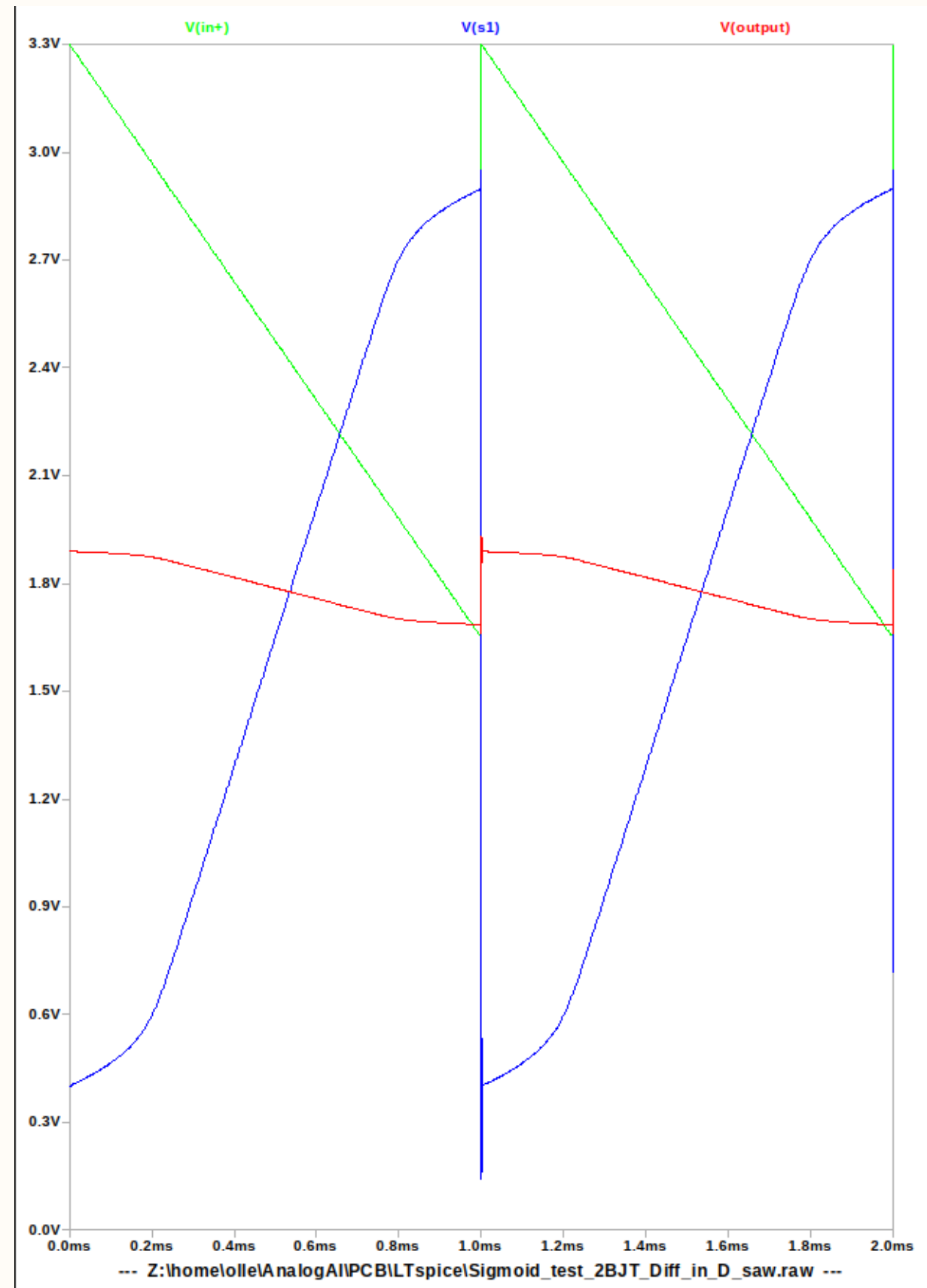
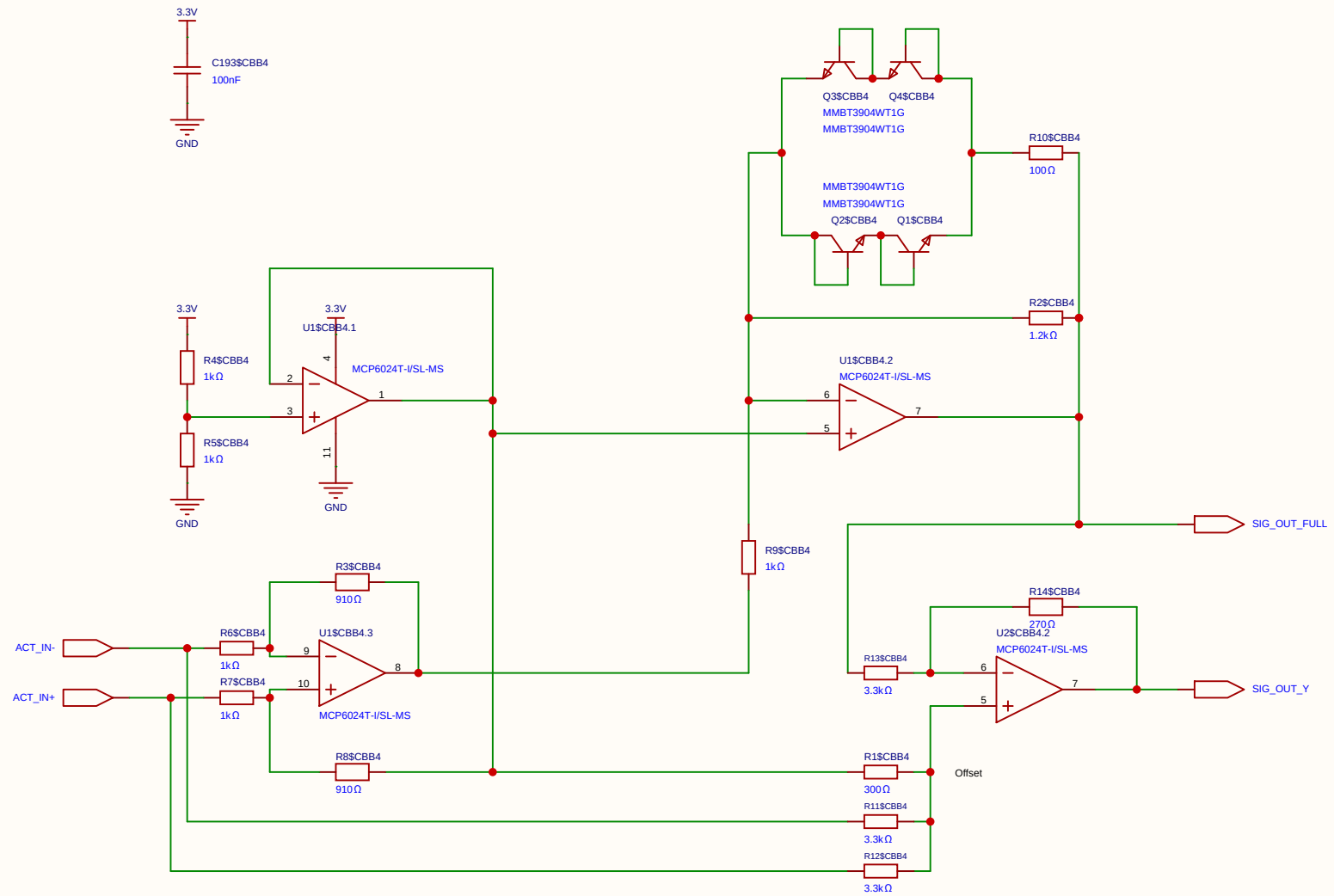
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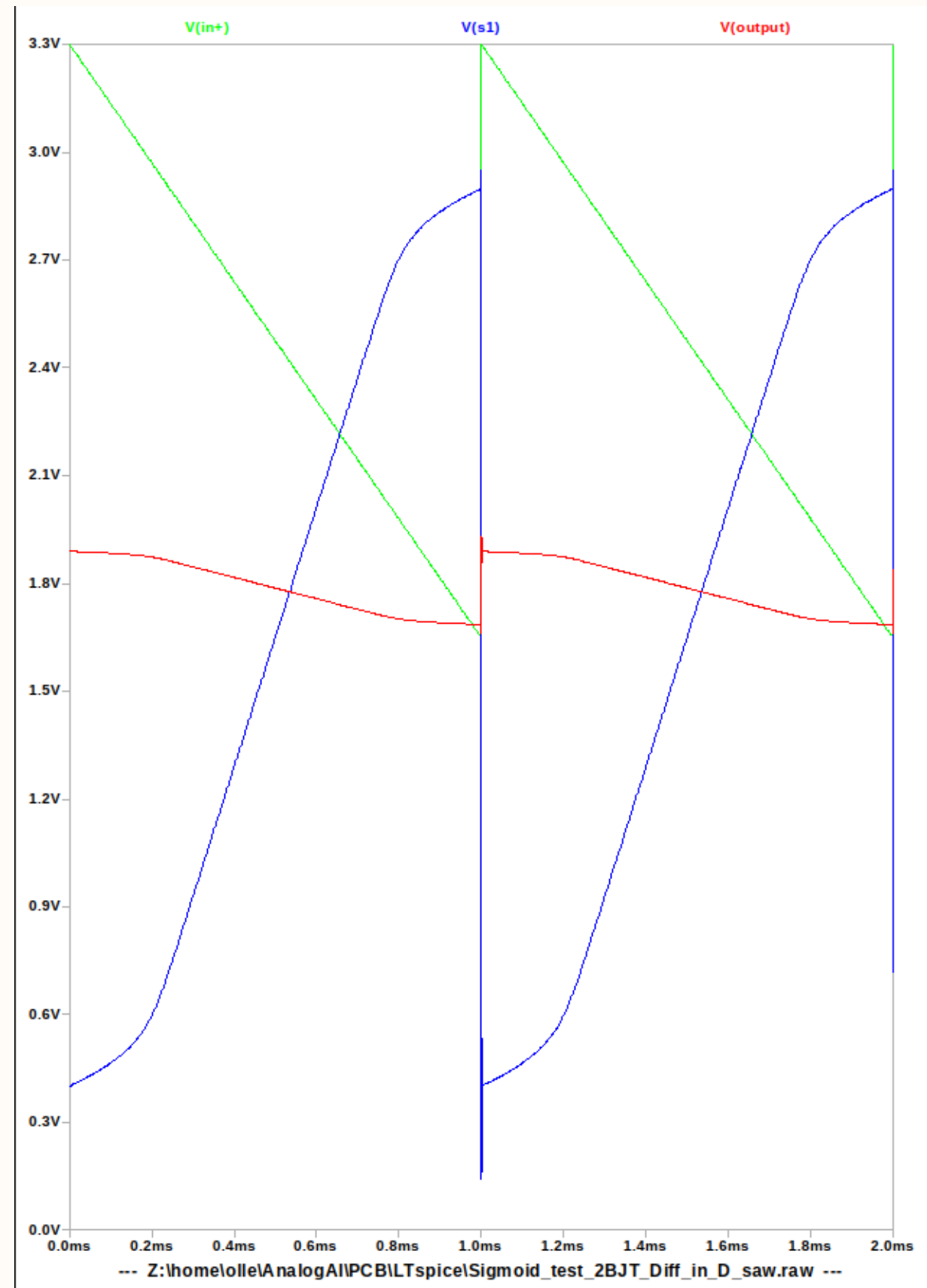
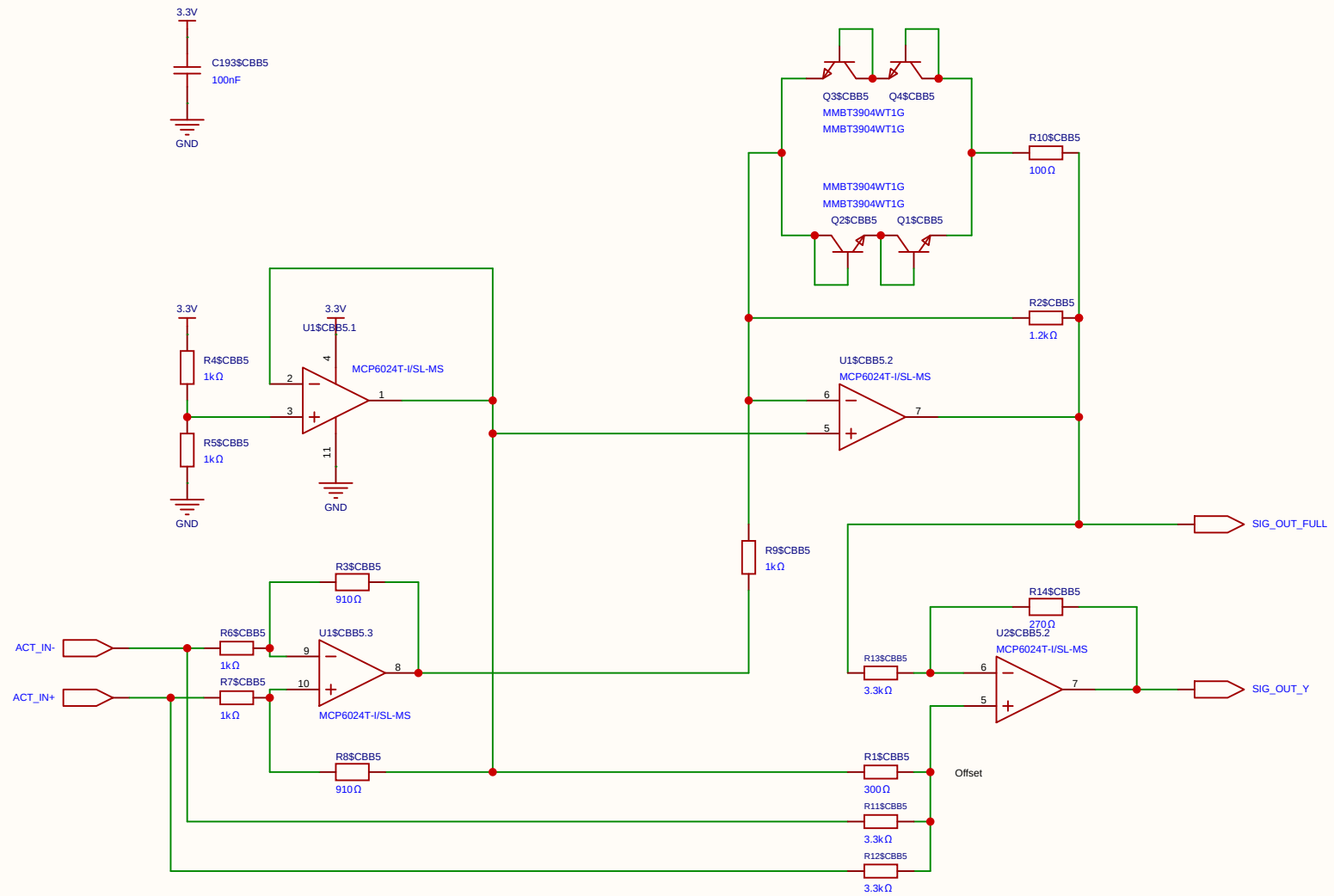
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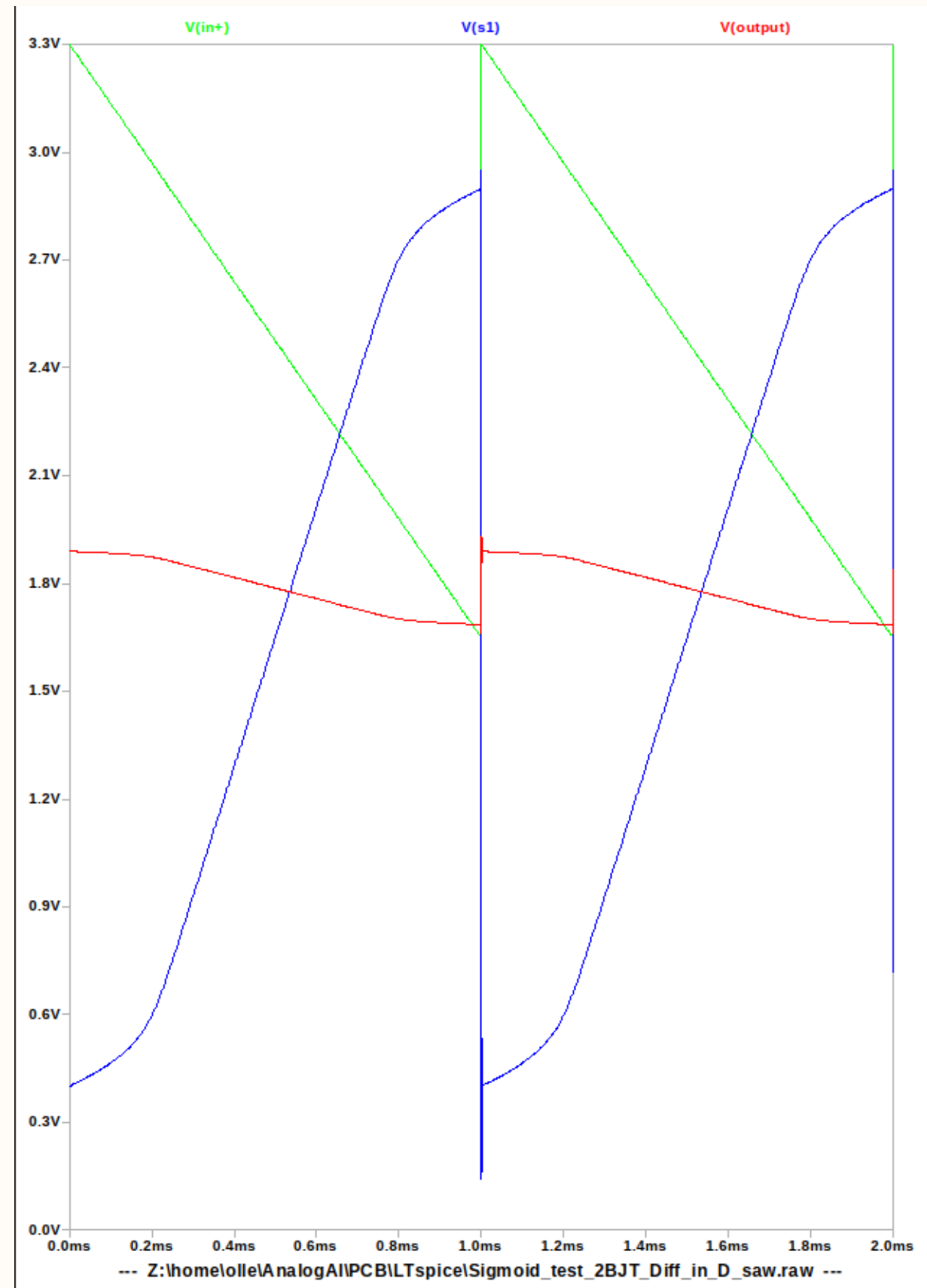
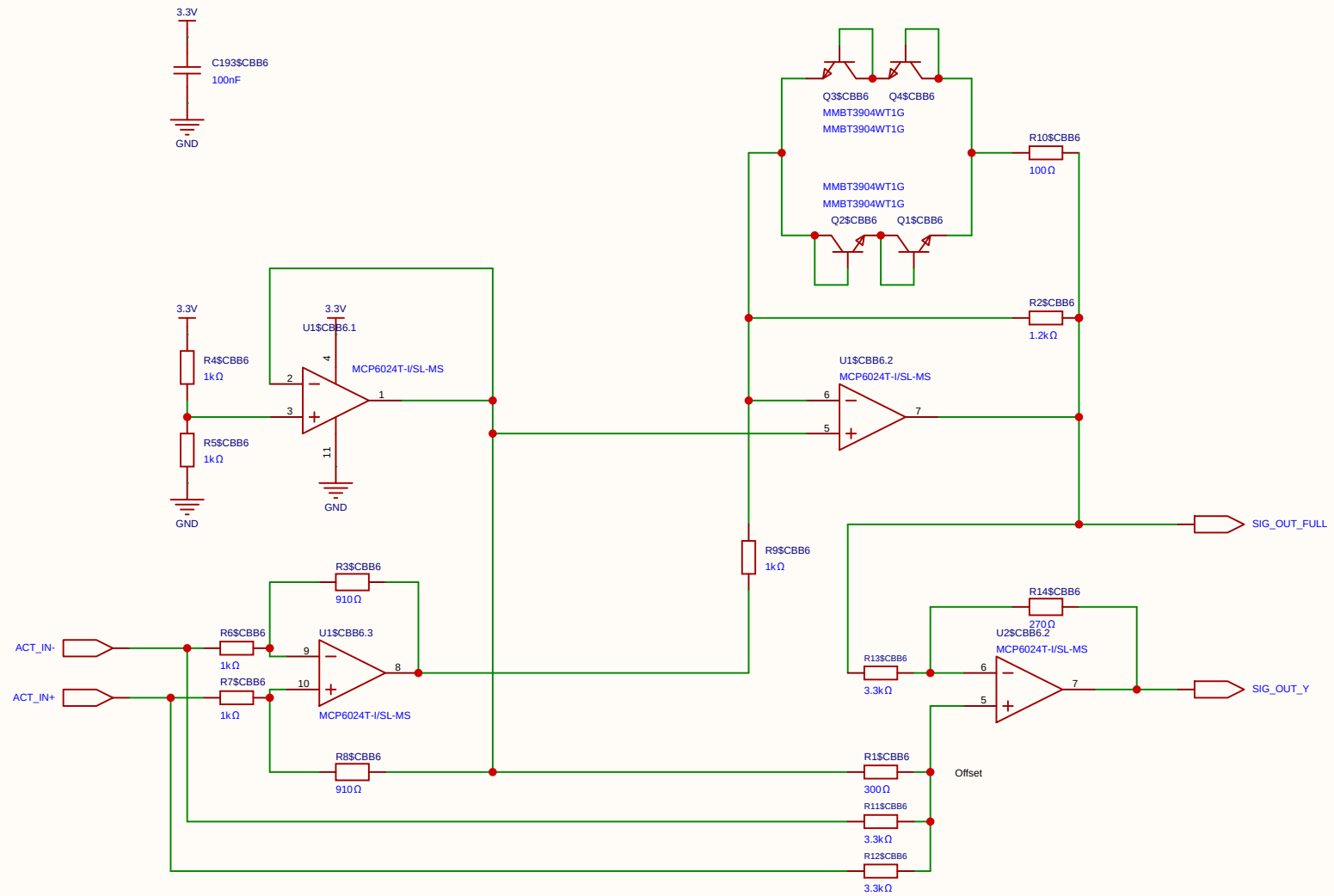
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Schematic	SIGMOID_CELL			Create at	2026-06-14
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